COMPARING THE PERFORMANCE OF COMMON TECHNICAL ANALYSIS INDICATORS USED IN CRYPTOCURRENCY TRADING

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ABSTRACT

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Comparing The Performance of Common Technical Analysis Indicators Used in Cryptocurrency Trading

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The main topic of this thesis is technical analysis indicators used in cryptocurrency markets for cryptocurrency trading. The thesis generally explores the theory behind technical analysis and the indicators used alongside with it, as well as compares the performance of the indicators using a method called backtesting for each individual indicator selected for the study. The chosen indicators in this study are the Bollinger bands (BB), simple moving average (SMA), exponential moving average (EMA), stochastic oscillator, and the relative strength index (RSI). The thesis also briefly examines the characteristics of Bitcoin and the operational principles of its blockchain, as the data used for backtesting is based on the price fluctuations of the Bitcoin cryptocurrency over a two-year period, from January 1, 2022, to January 1, 2024.

The results of the backtesting provided several different values that described the performance of the indicators. Among these values, the net-profit and the win-rate were chosen as the defining factors for determining the winner in the performance comparison. The net-profit is a reasonable choice because it indicates how much profit the indicator has generated during the evaluation period. The win-rate, on the other hand, defines the percentage of trades made by the indicator that were profitable for the trader.

Bollinger bands had the best net result among the indicators, so based solely on monetary efficiency, Bollinger bands emerged as the winner of the comparison. On the other hand, the study found that the RSI index had the best win-rate.

However, it is important to note that the study includes several factors that limit the application of its results to real trading. These include the simplicity of the indicators and the predefined time frame

TIIVISTELMÄ

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Yleisten kryptovaluutta kaupankäynnissä käytettyjen teknisten analyysi-indikaattorien suorituskyvyn vertailu

Kauppatieteiden kandidaatintutkielma

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65 sivua, 13 kuvaa ja 20 liitettä

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Avainsanat: kryptovaluutta, tekninen analyysi, kryptomarkkinat, Bitcoin, kryptovaluutat, kaupankäynti

Tämän kandidaatintutkielman pääaiheena ovat teknisen analyysin indikaattorit, joita hyödynnetään kryptovaluuttamarkkinoilla kryptovaluuttakaupankäynnissä. Tutkielmassa avataan yleisesti teknisen analyysin sekä siinä käytettyjen indikaattorien teoriataustaa sekä vertaillaan indikaattorien suorituskykyä hyödyntämällä menetelmää nimeltä taustatestaus jokaisen yksittäisen tutkielmaan valitun indikaattorin kohdalla. Valitut indikaattorit ovat tämän tutkielman tapauksessa Bollingerin nauhat (BB), yksinkertainen liukuva keskiarvo (SMA), eksponentiaalinen liukuva keskiarvo (EMA), stokastinen oskillaattori ja RSI-indikaattori. Tutkielmassa perehdytään myös pinnallisesti Bitcoinin ominaisuuksiin sekä sen lohkoketjun toimintaperiaatteeseen, koska pohjadatana taustatestaukselle toimi kryptovaluutta Bitcoinin kurssivaihtelut kahden vuoden suljetulla aikavälillä alkaen 1.1.2022 ja päättyen 1.1.2024.

Taustatestauksen tuloksena saatiin useita eri arvoja, jotka kuvasivat indikaattorien suorituskykyä. Näistä arvoista nettotulos sekä voittosuhde valittiin suorituskykyvertailun voittajan määrittäviksi tekijöiksi. Nettotulos on järkevä valinta, koska se määrittää kuinka paljon voittoa indikaattori on muodostanut tarkasteluaikavälillä. Voittosuhde puolestaan määrittää sen kuinka monta prosenttia tietyn indikaattorin tekemistä transaktioista on kohdistunut kaupankävijälle voittoisaksi.

Parhaan nettotuloksen indikaattoreista omasi Bollingerin nauhat, joten perustuen pelkkään rahamääräiseen tehokkuuteen Bollinger nauhat osoittautuivat vertailun voittajaksi. Toisaalta tutkimuksessa havaittiin, että parhaimman voittosuhteen omasi RSI-indeksi.

On huomioitavaa kuitenkin, että tutkimus sisältää monia tekijöitä, jotka rajoittavat sen tulosten liittämistä oikeaan kaupankäyntiin. Näitä ovat esimerkiksi indikaattorien yksinkertaisuus ja ennaltamääritelty aikaväli taustatestauksessa. Muokkaamalla esimerkiksi näitä tekijöitä on mahdollista saada erilaisia tutkimustuloksia liittyen tässä tutkielmassa käytettyihin indikaattoreihin

AI APPENDIX (TEKOÄLYN KÄYTTÖLIITE)

Artificial intelligence (Chat-GPT 4o mini) has been utilized in writing this thesis. AI has been used in order to summarize paragraphs. AI has also been utilized as a search tool in order to find suitable material for the theory section. AI has been also used to help with creating scripts for Pine Script. In order to validate the scripts, they were inserted into Pine Script editor in order to see if they provided estimated results. If the script was faulty Pine Script showed an error message which was used in order to make corrections. If the error message was not included the script was further edited in Chat-GPT in order to find and correct the possible error.

Appendix:

"find me relevant examples of articles on [insert topic]"

"summarize this claim [insert claim]"

"[insert script] [insert error message]"

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1 Introduction

The world is currently becoming more digitalized by the day and in recent years cryptocurrencies have become a relevant topic among investors as-well as with different institutional players. Technical analysis has played a part in asset trading since the early days of the stock market and its indicators can be regarded as important tools when it comes to predicting the price-behaviour of different assets. Because the cryptocurrency market behaves similarly in principle to the stock market, technical analysis can be therefore seen as a valuable method for analysing and predicting the price-behaviour of different cryptocurrency assets not limiting its usage to stock market assets only.

Compared to the fundamental analysis which is catered towards the factors involving analyses of a company's financial records which are for example the income statements, the balance sheets, or the cash flow reports, technical analysis focuses more on the immediate price movements, trends, and trading volume in order to be able to identify specific patterns and predict future market behaviour with different kinds of technical indicators.

There already exists a lot of academic research on technical analysis related to the stock market due to the fact that the stock market is a fairly old phenomenon by now. However, the rise of cryptocurrency markets has not gone unnoticed by the academic world, and papers discussing the implementation of technical analysis in cryptocurrency markets have surfaced over the years since the launch of the blockchain technology. This thesis will contribute to the growing size of research conducted on technical analysis and the cryptocurrency market by providing an example of how the methods and results of conducting technical analysis can be utilized in order to draw conclusions based on data gathered from real-world trading indexes.

In addition, to generate revenue by trading in the cryptocurrency market it is beneficial to be able make correct predictions of the price movements in the market and know the basics of the tools utilized in making these predictions. This thesis will also provide a practical overview of technical analysis, its most common indicators and how they can be implemented in the context of cryptocurrency trading.

1.1 Background

Before conducting the research, it is beneficial to discuss some of the concepts which have a significant role in this thesis. The following chapters 1.1.1 and 1.1.2 will include a brief description of the term cryptocurrency and blockchain which consistently appear in cryptocurrency related research and literature. Chapter 1.1.3 will provide a brief overview of the tools used for conducting technical analysis — TradingView and Pine Script.

1.1.1 Cryptocurrencies

The term cryptocurrency is an asset or an item of value, commonly referred to as a coin or a token which exists digitally, not physically and is generated by software. Cryptocurrency transactions exist within a system called the blockchain and are generated by the rules embedded within a code which creates and oversees them. Cryptocurrency as an asset has value and it can be exchanged for other types of cryptocurrencies or to different globally used currencies such as fiat-money. The value of cryptocurrency is often determined in marketplaces called exchanges which operate similar to a stock market. It is a place where the buyers and sellers of cryptocurrency come together online to trade their coins at a mutually agreed price-tag. The coins can be interpreted as unique items of data or strings of code which are then passed from account to account between the traders. These transactions are all recorded on their respective databases known as blockchains. (Lewis, A. 2018)

1.1.2 The Blockchain

The blockchain can be seen as a distributed digital ledger technology e.g. a database, which is maintained through its own network. Each member of the network, referred to as a node, verifies and accepts transactions within the blockchain by using technologies derived from different methods of computing and cryptography. This is done for the purpose of avoiding possible duplicated transactions (Bouveret & Haksar 2018). There are different types of blockchain technologies used for different cryptocurrencies.

After an agreement referred to as decentralized consensus is attained among the traders who participate in the network, the transaction is marked to the ledger and is then validated by the blockchain (Bouveret & Haksar 2018). The blockchain contains a history of transactions which is permanent and cannot be in theory manipulated in any way (Bouveret & Haksar 2018).

1.1.3 Trading View and Pine Script

TradingView is a popular web-based platform used for charting and analysing financial markets, including stocks, forex, cryptocurrencies, and other assets. It offers different price-charts which users can utilize to track and analyse price movements in real time. The platform supports various chart types such as candlestick, line, and bar charts, and provides numerous overlays, including indicators and drawing tools, to aid in for example technical analysis. Additionally, TradingView supports a wide array of built-in indicators, such as moving averages, enabling users to further simplify the process of analysing data and forming future price-predictions.

Pine Script is the scripting language used within TradingView to create custom indicators, strategies, and alerts. It allows users to define how an indicator behaves or to automate trading strategies. With Pine Script, users can create their own technical indicators, modify existing ones, or combine multiple indicators into a single custom tool. Pine Script also enables users to backtest trading strategies i.e. simulate the performance of an indicator or a strategy in retrospective. The backtesting feature uses the historical price-data of an asset stored on TradingView's website. Backtesting helps to assess the effectiveness of a strategy before applying it to live markets while Pine Script makes it simple for users to generate and backtest technical analysis indicators or strategies without the need for downloading price-data from an index or an exchange. This also negates the need migrate data to a separate programming software in order to conduct a backtest.

1.2 Research objectives

The goal of this thesis is to get a grasp of how technical analysis is being utilized in cryptocurrency trading by focusing on some of its common indicators which can be used by traders regardless of their initial experience in the market.

The performance of the indicators will also be compared in order to find out which of them had the best profitability alongside with other different factors such as win-rate. By examining different classic indicators such as oscillators, moving averages and Bollinger bands this thesis will provide a basic understanding about the indicators in theory as-well as provide an example of how to use these indicators in cryptocurrency trading, and how these indicators can be utilized in order to gather information that can be used in order to make future predictions of asset prices.

Research questions are the following:

Q1: How technical analysis can be utilized in cryptocurrency trading?

Q2: How have the common technical analysis indicators performed compared to each other in the Bitcoin cryptocurrency market?

The research will use historical data of Bitcoins price movements in order to backtest the different technical analysis indicators. The reason why Bitcoin's price-data was chosen for this research is due to Bitcoin being the most widely known and followed cryptocurrency in the market. The relevance of Bitcoin makes it a convenient asset to study and the methods and results of this research can also be utilized as a baseline for those who have an interest in conducting their own research on technical analysis and Bitcoin or other types of cryptocurrencies alike.

1.3 Conducting the research

The research will be divided into three distinct parts. The first part discusses Bitcoin and the technology behind it in order to summarize the technology as a whole before moving on to technical analysis and its indicators. The blockchain and cryptocurrencies were already discussed briefly in general, but the first part focuses only on Bitcoin and its blockchain. This works as a supplement for the thesis and also supports the fact that Bitcoin's price-data is being utilized in conducting the research.

The second part includes a detailed theory section about the common technical analysis indicators explaining them in broad detail, as-well as describing what technical analysis is and how it has been utilized by academic research in the past. The information about the indicators and technical analysis is gathered from different literature sources found online mainly through scientific articles, journals, and books regarding the topic. This part will help to form an understanding of what technical analysis and its indicators are and how they are related to cryptocurrencies.

The third part will show and explain how technical analysis is implemented in practice. The material for this part is cryptocurrency Bitcoin's historical price-data stored on TradingView. The reason why TradingView was chosen as the platform to gather the data and the results for this study is due to its convenience, popularity and ability to backtest the chosen indicators for this research without the complexity of external data-management as stated before. This part contains aspects of a quantitative research. According to Coghlan & Bryndom-Miller (2014) the use of quantitative research methods makes it possible to conduct different types of statistical analyses which can range from simple to extremely complicated.

2. Bitcoin and its Blockchain

Prior to this chapter cryptocurrencies and blockchain technology were mentioned briefly. This chapter will go more in depth regarding Bitcoin and its own blockchain technology discussing the technology in further detail including the aspects of mining and decentralization. The blockchain technology and Bitcoin can be considered revolutionary having many benefits including fast and secure transactions of value between its users. However, it also has some negative sides to it that cannot be discarded. This chapter will also discuss some of the negative aspects when it comes to Bitcoin and its blockchain.

2.1 Overview of the Bitcoin Blockchain

Bitcoin can be represented as transactions which are recorded in its own blockchain. The blockchain acts like a global spreadsheet of transactions. The blockchain also uses Bitcoin's peer-to-peer network system while verifying and approving transactions. The bitcoin blockchain is considered distributed which means that it is run on computers provided by unique users i.e. nodes around the world. The blockchain is encrypted in order to make it secure. (Tapscott & Tapscott 2016, p 23)

Furthermore, when it comes to security there exists no singular databases which could be hacked and the blockchain remains also public which means that anyone can see it whenever they want to due to it existing on a network shared by its users and not being owned by a separate private institution (Tapscott & Tapscott 2016, p 23). This means that stealing Bitcoin in the traditional sense is impossible. If one would want to steal Bitcoin it would require sophisticated hacking methods or social engineering.

The process of running the Bitcoin network involves several procedures. The process begins by initiating transactions which are broadcasted to other nodes currently running the network. Then each node will begin the process of collecting transactions into a block while at the same time working together on finding a proof-of-work for that block. This is regarded as a highly difficult process computationally. Once a node manages to find a proof-of-work, it will show the block containing the proof-of-work to all the other nodes. The other nodes will then proceed to accept the block but only if all transactions within the block are valid and have not been previously spent. After accepting the

block, nodes indicate their acceptance by starting to work on generating the next block following the accepted block within the chain repeating the process from the start. (Nakamoto 2008)

The way the Bitcoin blockchain runs the pending transactions is through a process called hashing. During the hashing process the blockchain utilizes a secure hash algorithm called the SHA-256 in order to validate and solve blocks (Tapscott & Tapscott 2016, p 235). The point of a hashing algorithm is basically to act as lock which secures the transactions making it sure that they cannot be tampered or manipulated. In practice, hashing algorithm takes the transaction and turns it into a string which looks like random numbers and characters combined.

Bitcoin achieves proof-of-work by applying the use of entities referred to as hash puzzles (Narayanan et al. 2016, p 64). They can also be viewed as complex mathematical equations in a more generalized term. The main objective of a proof-of-work system is to approximate a selection of a random node in proportion to a resource which cannot be easily monopolized. In the case of blockchains, that resource is computing power which can be seen as the main factor that makes a blockchain network a proof-of-work system. (Narayanan et al. 2016, p 64) The nodes connected to the network compete with one another using computing power in order to solve the hash puzzles. This results in nodes with more computing power being more likely to be picked by the system for receiving a reward due to the proportion of their computing power being greater compared to lesser nodes (Narayanan et al. p 64).

In order to generate a block, the node which advocates a block is also demanded to discover an entity referred to as a nonce (Narayanan et al. 2016, p 64). In cryptography, the term nonce is defined as a value that can only be used once (Narayanan et al. 2016, p 23).

The bitcoin blockchain updates every ten minutes. During this time all the transaction conducted are verified and stored on the block which is then linked to the previously generated block thus creating a chain of blocks, hence the name blockchain. Each block must also root to the previous block to be considered as valid. In more simple terms the system generates indefinite timestamps which store exchanges of value, thus preventing attempts to alter the ledger. For a person to modify or alter the ledger it would take the effort to rewrite the code of the whole blockchain making it computationally impossible. (Tapscott & Tapscott 2016, p 23)

Tapscott & Tapscott (2016, p 23) summarize blockchain technology as a distributed world-wide ledger which anyone can download and run on their computer similar to the World Wide Web. The blockchain technology behind Bitcoin allows reconciliation of digital records of almost any kind in real-time (Tapscott & Tapscott 2016, p 23).

2.2 Mining process

Bitcoin is generated through mining which can be seen as a computation process executed by a miner. In order to start mining Bitcoin, the miner needs to have a connection with other nodes residing in the network. Once connected there are a couple of basic tasks according to Narayanan et al. (2019) that the miner will perform to generate new Bitcoin into circulation.

In the start of the mining process the miner searches for open transactions and validates them by checking that their authenticity matches and making sure that the outputs spent currently have not been spent in the past. Second part of the process is to keep the blockchain running and searching for blocks. The blockchain must be maintained constantly. The miner connects with other nodes giving them the historical data of blocks that are already part of the blockchain prior to the miner joining the network and starting the process. Then the miner starts the search for new blocks broadcasted to the network. The miner needs to validate each block that they find by confirming each transaction in the block and checking that the block contains a valid nonce. (Narayanan et al. 2016, p 131)

The miner constructs a candidate block and once they have an updated copy of the blockchain, they can start assembling their own blocks. In other words, the miner will group transactions that they have found about into a fresh block that expands to the latest block they have information about. Each transaction included in the block must be confirmed as valid. By finding a nonce the miner confirms that the block is valid. This step requires a lot of resources computationally in order to succeed in, and it is where many miners face a lot of difficulty. (Narayanan et al. 2016, p 131)

When the block is accepted by each miner in the blockchain the miner receives a reward. The reward also includes a transaction fee which is subtracted from the initial amount the miner receives. (Narayanan et al. 2016, p 131)

2.3 Decentralization

There have been numerous failed attempts when it comes to digital or electronic cash before the advent of Bitcoin. Bitcoin’s main difference compared to most of failed digital cash system is its decentralization-aspect. The driving factor of Bitcoin that enables decentralization is its blockchain technology. (Narayanan et al. 2016, p 293) When it comes to decentralization it basically means the network of computers work peer-to-peer between each other and do not rely on one central server or a database (Walch 2019).

The second way that decentralization is commonly referred to according to Walch (2019) is that it is used to explain how agency behaves within the blockchain. Since there is no single party keeping the record of the transactions no single party can be held accountable for them. In a system like the Bitcoin blockchain accountability could be seen as everywhere and nowhere simultaneously. (Walch 2019)

The fact that Bitcoin works as a decentralized currency devoid of control from a single party or an entity such as a central bank is one of the common factors often mentioned as a positive aspect of the technology. It can, however, have negative sides to it as-well. When nobody has control over how Bitcoin is used it can be utilized for malicious purposes for example money laundering or other illicit activities.

2.4 Issues with Bitcoin

Bitcoin can still, in the year 2024, be seen as a relatively new asset and a store of value, which is still in a developmental phase compared to other more traditional methods of investing such as futures or stocks. This leads to unique problems such as regulatory issues and market risks which can hold Bitcoin back from being adopted as a generally used transaction method. There is also the fact that Bitcoin uses a lot of energy to keep the blockchain process running which can create issues with the environment for instance. There is also the problem of Bitcoin being a highly volatile asset which might discourage investors who value balance and consistency in their portfolios. It is also worth noting that regardless of the encryption of the blockchain, Bitcoin is still susceptible to different kinds of security risks. The following chapters will discuss these problems that Bitcoin faces in further detail.

2.4.1 Adaptation

Bitcoin represents a challenge to the traditional financial system and to the government control over monetary policy. Bitcoin as a currency function externally to the traditional financial institutions such as banks and without borders. Therefore, its users have been able to avoid the existing administrative structures (Hendrickson et al. 2015, p 5). Transacting with Bitcoin excludes the user from the typical disclosures or reporting utilized in banking activities, as-well as having no inquiries on transactions (Hendrickson et al. 2015, p 5).

Because of these features Bitcoin is easy to use for illicit transactions. Governments are concerned that its use could facilitate illegal activities and disrupt key aspects of government policy, such as monetary control and revenue collection (Hendrickson et al. 2015, p 20). In response, several governments have attempted to restrict or outright ban the usage of Bitcoin (Hendrickson et al. 2015, p 20). This has been a factor that has disrupted the adaptability of Bitcoin as a mainstream currency.

2.4.2 Energy consumption

Bitcoins mining resources have raised various environmental concerns. Due to the rapid growth in the popularity of the currency the amount of energy needed to validate the transactions and operate the blockchain network have grown on the side (Siddique et al. 2023). The rising energy requirement is mainly due to the proof-of-work consensus which are the difficult-to-compute mathematical calculations done with high-power computer systems resulting in growth in the environmental effect of mining (Siddique et al. 2023). Furthermore, according to Siddique et al. (2023) Bitcoin relies mainly on the non-renewable energy sources for its electricity requirements resulting in carbon emissions and issues with the long-term viability of the blockchain operation.

2.4.3 High volatility

According to Baur & Dimpfl (2021) Bitcoin cannot be utilized like traditional fiat currencies due to Bitcoin's high volatility. The volatility is an issue because it makes Bitcoin notably expensive to be used as an exchange method or a storage of value (Baur & Dimpf 2021). Furthermore, the high volatility also affects Bitcoin's role as an investment, because according to Baur & Dimpfl (2021) assets with exorbitant volatility levels do not tend be solid in diversifying the risks of a portfolio.

2.4.4 Security

Because of Bitcoin's decentralized model it can be classified as an uncontrollable environment which makes subject to different types of attacks from hackers or thieves. A variety of different techniques have been developed in order to disrupt the operation of blockchain. In this chapter some of the most famous attacks on Bitcoin's infrastructure will be briefly discussed.

One of the most notable breaches occurred with Mt. Gox, the largest Bitcoin exchange, where hackers stole 744,408 bitcoins, leading to its closure and a drop in Bitcoin's value. Other incidents include attacks on Silk Road (in 2013), where a vulnerability in the Bitcoin protocol was exploited, and Sheep Marketplace, where 5400 bitcoins were stolen. While the legitimacy of these attacks is debated, some suggest internal fraud rather than external hacking. (Conti & al. 2017)

In 2015, Bitstamp, an alternative exchange, was hacked due to a phishing attack, resulting in the theft of $5 million, but customer funds were not lost. Poloniex also suffered a hack in 2014, losing 12.3% of its bitcoins. In 2016, BitFinex experienced a major breach where 120,000 bitcoins were stolen due to a wallet vulnerability (Conti & al. 2017).

These incidents highlight the vulnerability of cryptocurrency exchanges and the increasing targeting of digital wallets as Bitcoin's value rises. Even though Bitcoin has extensive security measures it is still susceptible to attacks or tampering. This is a key factor that traders and investors in the cryptocurrency space should consider.

Furthermore, according to Conti et al. (2017) exchanges require highly experienced developers to secure Bitcoin effectively, stressing the importance of accountability and robust security measures to protect users from potential threats.

3 Technical Analysis

This chapter includes a basic overview of technical analysis and what it is in practicality as-well as highlighting some of the advantages and disadvantages of it. The chapter will explain in detail what technical analysis is based on and what kind of factors are involved when conducting technical analysis. There will be also some history about the performance of technical analysis methods and strategies from the research of real-world markets. This chapter will not go into the theory of the indicators used in technical analysis. The purpose of this chapter is to provide a basic understanding of technical analysis including the positive and negative aspects of it.

3.1 Overview of technical analysis

Technical analysis involves analysing market data, focusing especially on price movements. This information is then utilized in order to recognize possible trading windows or opportunities in the future. Technical analysis is based on the so called efficient-market-hypothesis which states that all of the important market information is displayed in price patterns and charts (Malkiel 2003; Grunder et al. 2010). Technical analysis methods focus also on trends, momentum, and recurring patterns, which are repeated over time and can be also linked to human behaviour and market psychology (Grunder et al. 2010). The market charts can therefore be seen in a more abstract manner as a mirror of the collective hivemind of all the market participants instead of forming from fundamental economic factors alone (Grunder et al. 2010).

Technical analysis can be also viewed as a collection of prediction methods such as chart and cycle analyses or trading systems utilizing computational methods (Irwin & Park 2007). A trading system is formed from a set of trading rules which exist to generate signals for trades (for instance: long, short, or out of the market) based on different values of different parameters set by the user (Irwin & Park 2007). Different trading systems incorporate common technical indicators which include moving averages, channels, and momentum oscillators (Irwin & Park 2007). All of the relevant indicators included in this thesis will be explored further in theory during later chapters.

It must be noted however that technical analysis in on itself is not a magical solution for beating the market. Instead, it offers the trader or the investor a set of tools which can help with acquiring financial success when it comes to trading assets. To support this claim, Irwin and Park (2007) found

out that among a total of 95 modern studies, 56 studies uncovered promising results when it comes to the utilization of technical indicators. It is also noted that 20 studies contained unfavourable results, and 19 studies in total indicated results in-between good and bad values (Irwin & Park 2007). Despite a clear existence of evidence favouring the technical trading indicators, Irwin and Park (2007) also argue that empirical studies can be limited to multiple issues in when it comes to the testing methods. A few examples of these issues are mistreatment of data, ex post selection of trading rules or search technologies, and difficulties in the estimation of transaction fees or the risks related to trading (Irwin & Park 2007). This can further skew the overall perceived success rate of different technical analysis methods used in buying or selling tradable assets.

It is also stated that high volatility and past losses in value have been associated to have a significant beneficial effect when it comes to the profitability of trading strategies which utilize technical analysis (Parvez et al. 2000). An example of this kind market atmosphere would be according to McKenzie (2007) the Asian currency crisis of 1997, which established the onset for widespread economic chaos amongst the emerging markets sectors. This led to remarkable falls in share values and billions of dollars lost from market capitalization. The general disposition of market movements has been associated to have a notable effect on the outcome of trading strategies utilizing technical analysis (McKenzie 2007). Cryptocurrencies such as Bitcoin can be regarded as highly volatile assets which can radically lose or gain value in a short time-period. Therefore, the historical success of technical analysis methods used in volatile and downtrend markets such as the Asian currency crisis of 1997 could also be linked to their widespread popularity in the cryptocurrency market of today.

In summary, technical analysis methods and its indicators can be utilized in practice to form a more comprehensive understanding of the pricing of the asset. It offers investors a more objective approach in predicting the value of an asset. Technical analysis methods help investors to make decisions based on numerical data and to form a prediction of future price behaviour. Without this investing could be seen more as pure guessing on how the price of the asset will behave. It should be noted that this conclusion does not include the possibility that the trader has insider information on the asset which could be used to enhance correct decision making in order to profit from the market.

3.2 Advantages and disadvantages of technical analysis

In the previous chapter (3.1) there was a mention about the fact that technical analysis should not be seen as a crystal ball technique which can be used to correctly predict the price of an asset. It is also confirmed that studies conducted on the profitability and reliability of technical analysis have also

shown negative and mixed results. Even the said results from the studies have a possibility to suffer from issues when it comes to their actual reliability. However, using technical analysis carries many positive aspects to it as-well. This chapter will describe the advantages and disadvantages of technical analysis in detail and discuss the possible reasons why technical analysis might fail in predicting future price movements and misguide the trader or investor to making incorrect decisions while also highlighting some of the positive advantages.

One of the key advantages when it comes to technical analysis is the fact that ignores all the possible economic, technological and other factors when it comes to the industry or company that is tied to the asset. Technical analysis focuses purely on data on the prices and the trading volume in order to make estimations on the future price of the asset (Petrusheva & Jordanoski 2016). The trader will assume that the emerging pattern which has been formed by using technical analysis will also repeat again in the future (Petrusheva & Jordanoski 2016). According to Jarunde (2023) technical analysis helps investors with determining the correct market sentiment which provides insights to prevailing market trends and the overall psychology of buyers. Technical analysis also helps the trader or the investor in timing trades i.e. identifying key exit and entry points based on chart patterns & indicators and adapting to the to the underlying market conditions which allows for quick adjustment to changing market dynamics (Jarunde 2023). However, there can be a downside when it comes to relying purely on the results. Markets tend to act randomly, and relying too much on the results provided by technical analysis has a possibility to lead the investor or the trader astray disconnecting them from other methods of market analysis. According to Jarunde (2023) technical analysis methods can also give false or misleading signals or even break down especially in volatile markets. This is extremely important to keep in mind especially when trading cryptocurrency since it is notoriously a heavily volatile asset class.

Petrusheva & Jordanoski (2016) state that nowadays there are a lot of computer programs and tools which help conducting technical analysis and making it more accessible for an average investor or a trader. Unlike fundamental analysis which requires substantial research or knowledge about the asset or company, such as the potential of future growth or in the case of cryptocurrency the possible utilization of the token, technical analysis can be conducted simply by entering the parameters required to the corresponding computer program in order to start receiving investment signals (Petrusheva & Jordanoski 2016). However, this also requires the understanding of said signals and basic financial knowledge to execute the correct moves. If the trader or the investor has no knowledge on how to act based on the results gained from technical analysis, then there is a possibility to make an incorrect or unwanted decision such as selling the asset when it should have been held onto.

According to Jarunde (2023) technical analysis methods are also universally applicable. They work on a variety of different asset classes such as stocks, commodities, and currencies (Jarunde 2023). This also includes cryptocurrencies. This can be seen as a major benefit since the trader can learn each method only once and it can then be used in multiple different markets which saves time and makes it possible to focus on other aspects of trading such as fundamental analysis or market research.

Another advantage of technical analysis is also the fact that it does not take into account the subjective aspects of a certain asset or company such as the trader's or analyst's personal expectations (Petrusheva & Jordanoski 2016). The decision which the trader makes is based purely on results from the analysis which are tied to the signal given by the indicator (Petrusheva & Jordanoski 2016). This basically means that the trader or the analyst does not consider his or her feelings about the asset or the company. For example, if the company represents something that does not align with the current ideology or trend of the market in general, the trader might think this will negatively affect the future value of the asset. Relying on the results of technical analysis basically neglects this possibility completely.

On the contrary, it must be noted that subjective aspects can also hold value and importance to the trader. It is obvious that profiting from investments and trading is not completely a one-sided numbers game. Factors such as other players in the market and their emotions and feelings about a certain asset can significantly affect the price-behaviour of an asset because it is the investor-money that gives the asset its value in the end. In today's world social media also plays a big part when it comes to investing especially when it comes to the cryptocurrency market. If the trader relies only on numbers and graphs but neglects the expectations from other investors, which are voiced for example on different social media platforms, this can turn into a massive disadvantage, since in this case the trader ignores the real world completely. In addition, Jarunde (2023) states that technical analysis methods have a downside of neglecting fundamentals i.e. using them can make the trader or the investor overlook the assets long-term prospects or the company's underlying financial health.

The disadvantage of technical analysis according to Petrusheva & Jordanoski (2016) is that lacks any significant academic or scientific authentication and that it cannot be validated by any rational arguments. The critics of technical analysis say that the methods are more based on intuition rather than actualized facts or solid evidence because a chart formation or an indicator in on itself cannot state with certainty that a stock or an asset price will behave in manner predicted by them (Petrusheva & Jordanoski 2016). However, if enough enough investors believe in the power of the indicator, it will cause them to act in a certain manner, either buying or selling the asset, which in turn will make the prediction made by the indicator become true (Petrusheva & Jordanoski 2016). If this happens

enough it will naturally make the traders and investors more confident about the performance of technical analysis methods and their indicators which in turn boosts the credibility of said methods and indicators. However, according to Jarunde (2023) this can be also seen as a downside since it makes the technical analysis methods act as self-fulfilling prophecies causing the market to react in the way of the indicator instead of real world. There is also the chance that the interpretation of chart patterns can vary between their users which leads to potential divergent signals when it comes to buying or selling the asset (Jarunde 2023).

4 The Indicators Behind Technical Analysis

Technical analysis methods and its indicators have been in use for example in stock trading for decades, therefore they are not new. Cryptocurrency market & cryptocurrency trading are however recent phenomena. Furthermore, as stated in the previous chapter's sub-chapter (3.2) the methods and indicators of technical analysis are applicable to different forms of asset-pricing-analysis. This basically means that the same methods used in analysing stock price-behaviour can be utilized in predicting cryptocurrency price movements.

The list of different indicators used in technical analysis is vast and it is not possible to include them all due to the scope of this research. Therefore, only a few were chosen, them being the Bollinger bands, simple moving average (SMA), exponential moving average (EMA), stochastic oscillator and relative strength index (RSI). These indicators were chosen because they are widely used within the cryptocurrency trading community as-well as simple to understand for someone who has no experience in trading while utilizing technical analysis. This chapter will explore the theory behind the indicators in detail in order to gain a basic understanding on how they function and how they should be used in trading.

4.1 Bollinger Bands

During the 1970s and 1980s John Bollinger realized the importance of focusing on volatility as key feature in index option trading in order to form his own approach to trading bands (Bollinger 1992). Trading bands are the lines which are constructed around a price graph of an asset in order to form an envelope. The movement which occurs close to the edges of the envelope is the key point of interest (Bollinger 1992). Bollinger bands are constructed by plotting two standard deviations around a simple moving average. Figure 1 shows the mathematical formula for the calculations as well as the trading bands which the Bollinger bands is originally based on.

In practice, moving standard deviations are applied to generate bands around a moving average. The price information used to calculate the standard deviations is identifcal to the information used to form the simple moving average (SMA) in-between the standard deviations. The calculations are done over a timeframe which has an intermediate-term trend. The central band is the n-day moving

average. The band above the SMA is the n-day moving average plus two times the standard deviation from that average, while the band below the SMA is conversly the n-day moving average minus two the deviation. The letter n represents the time-frame which is selected to reflect the intermediate-term trend. (Bollinger 1992)

Typically, when the price crosses the upper band, it signals that the asset is overbought, which is supposed to be interpreted as a selling signal. Contrarily, when the price crosses below the lower band, it indicates that the asset is oversold, which is supposed to be interpreted as a buying signal. (Windasari et al. 2018)

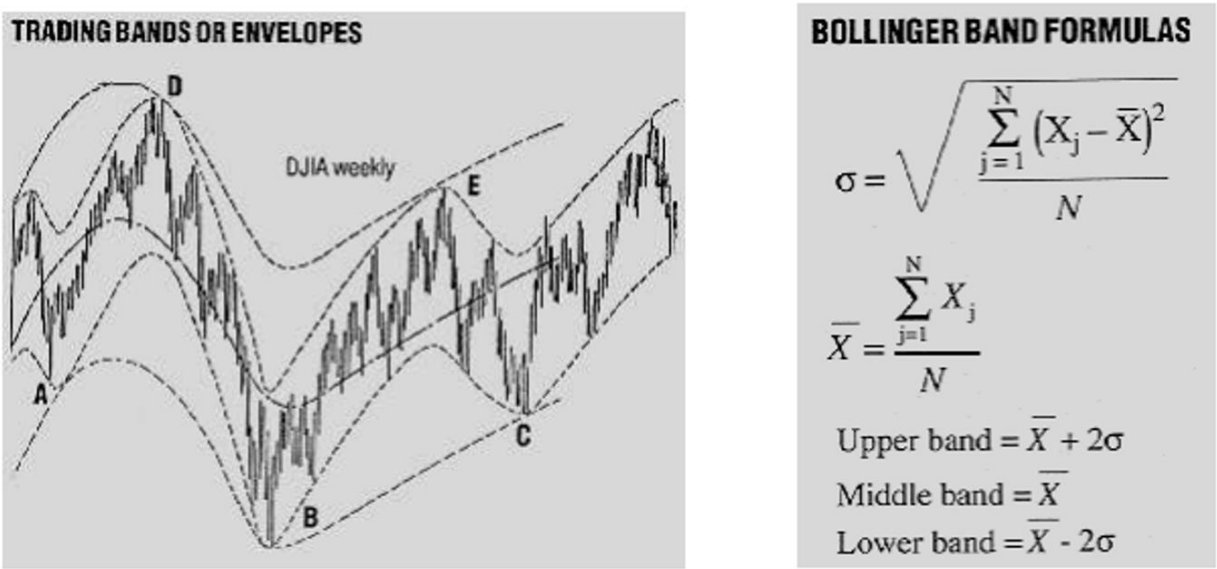

Figure 1. The trading bands referred to as envelopes on the right & the formula for Bollinger Bands on the left (Bollinger 1992)

Figure 2 illustrates the appearance of Bollinger Bands on a graph, where the moving average is a 20-day simple moving average, and the bands are shifted up and down by 4 %. According to Bollinger (1992), the bands respond very quickly to significant market movements.

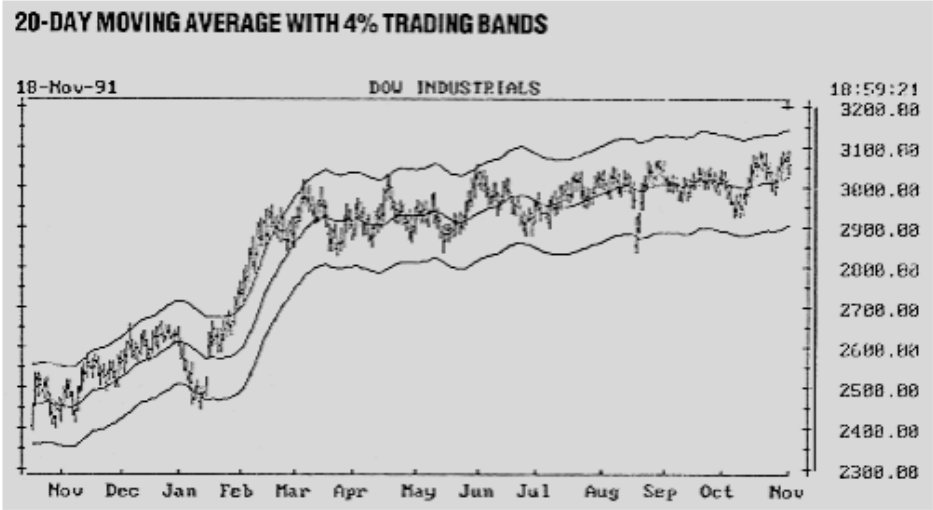

Figure 2. 20-day moving average with Bollinger Bands (Bollinger 1992)

4.2 Moving averages

Moving averages utilized in order to identify trends in asset pricing by smoothening the fluctuations of prices which makes it easier for the trader or the investor to observe information from the data (Ellis & Parbery 2005). Moving averages are considered lagging indicators, because they display the average price of an asset over a timeframe that has already passed (Ellis & Parbery 2005). Multiple different versions of moving averages have been developed for technical analysis, but the main principle stays the same. In this study only the simple moving average (SMA) and the exponential moving average (EMA) are chosen for further examination. Buy and sell signals for moving averages are usually given when the price crosses the moving average, the moving average changes direction or the moving averages cross each other (Mazumder et al. 2023).

Shorter period moving averages respond more quickly to changes in asset prices but are more susceptible to false signals due to small, unpredictable price fluctuations, leading to potential losses. Conversely, longer period moving averages diminish the impact of minor price movements, focusing on more significant trend. Longer period moving averages tend to also provide fewer trading signals which are more delayed compared to shorter averages. Regardless of the timeframe, moving averages generally fail to succeed in markets, where a clear trend is not evident. (Ellis & Parbery 2005)

Originally, the desire to make better use of historical data has driven the adoption of the moving average. A moving average with n = 5 observations simply shows the average of the five most recent data points (Hunter 1986). As each new observation is added, the oldest one in the set is being

removed from the equation (Hunter 1986). Moving averages can generate plots that exhibit cyclical and trend-like patterns, even when the original data consists of independent random events with a fixed mean (Hunter 1986).

In the following chapters 4.2.1 and 4.2.2, two different moving averages will be explored in further detail, them being the simple moving average (SMA) and the exponential moving average (EMA).

4.2.1 Simple moving average (SMA)

According to Ellis & Parbery (2005) the simple moving average is the most commonly adopted of all the technical indicators. However, it must be noted that this claim by Ellis & Parbery was made in 2005 which is almost 20 years ago. Nevertheless, it is a well-known fact that traders still utilize the SMA as an indicator in technical trading today so its popularity can be regarded as high. The mathematical formula of SMA is represented in figure 3 where n is the number of periods included within the average and k is the relative position of the period currently being considered within the total number of periods. P_t is the asset price at time t.

$$\mathrm{SMA}_n = \frac{1}{n} \sum_{t=k-n+1}^{k} P_t$$

Figure 3. The formula of simple moving average (Ellis & Parbery 2005)

Each price in the simple moving average (SMA) is given equal weight. Assuming that older prices are just as relevant as newer ones, can cause the SMA to become unstable, however this effect lessens as the duration of the simple moving average increases (Ellis & Parbery 2005).

A simple moving average is based on historical data; therefore, it does not predict future market movements. Instead, it trails behind the prevailing price. In a bull market i.e. an uptrend market, the lag effect places the SMA line below the price line, while in a bear market i.e. a downtrend market, it appears above the price line. When the market changes direction shifting from an upward to a downward trend or the other way around the SMA and the price line will cross each other. This happens because the SMA, due to its lag, continues to reflect a previous market trend. The point where the lines cross serves as a key signal in most moving average-based strategies. The basic trading rules derived from these crossings are the following:

1) A buy signal occurs when the current price rises above the moving average.
2) A sell signal occurs when the current price falls below the moving average.

4.2.2 Exponential moving average (EMA)

The SMA-indicator has been criticized due to the fact that every single price point in the series which is being averaged is equally weighted. Because of this the SMA-indicator assumes that older prices are as relevant as newer ones (El-Khodary 2009). This notation led to the development of various types of newer moving averages which give more weight to recent prices due to the belief that recent prices have a larger influence on expected results (El-Khodary 2009). This led to the adoption of the exponential moving average. According to El-Khodary (2009) the exponential moving average gained ground rapidly and quickly surpassed the simple moving average in popularity.

According to Hunter (1986) the main characteristic of the EMA-indicator which sets it apart from the SMA-indicator is that it gives less weight to data points as they get older along the timeframe set for the equation. The EMA-indicator is calculated by adding a percentage of the previous moving average to a percentage of the current price value. The mathematical formula for the EMA-indicator is as follows:

$$\mathrm{EWMA} = \hat{y}_{t+1} = \hat{y}_t + \lambda e_t$$

$$= \hat{y}_t + \lambda(y_t - \hat{y}_t)$$

where

$\hat{y}_{t+1}$ = predicted value at time $t + 1$ (the new EWMA)
y_t = observed value at time t
$\hat{y}_t$ = predicted value at time t (the old EWMA)
$e_t = y_t - \hat{y}_t$ = observed error at time t
and λ is a constant $(0 < \lambda < 1)$ that determines the depth of memory of the EWMA.

Figure 4. The formula of exponential moving average along with the explanation for symbols (Hunter 1986)

To add further clarification, EMA does not drop out the values of previous days. Instead, the influence of the previous values exponentially decreases and as time moves on, they become basically non-existent. Figure 6 shows the same formula as figure 5 but with a modernized look using symbols more common in today's econometrics. In figure 5 the P_k is the price of the asset, k is the relative position of the period currently being considered and α is the so-called "smoothing factor" calculated by 2/(1+n). The smoothing factor α is a constant between 0 and 1.

$$EMA_k = \alpha P_k + EMA_{k-1}(1-\alpha)$$

Figure 5. EMA-indicator formula with alternative symbols (El-Khodary 2009)

The buy- and sell-signals are the generally the same as with the SMA-indicator and all the other moving averages:

1) A buy signal occurs when the current price rises above the moving average.
2) A sell signal occurs when the current price falls below the moving average

4.3 Oscillators

Oscillators are a popular and widely used tool in technical analysis because of an attribute called the leading signal generating ability (Naved & Srivastava 2015). Leading signal generation means that the indicator does not lag behind the price action which sets them apart from moving averages (Naved & Srivastava 2015). Oscillators are utilized in sideways market while as the previous indicators such as the moving averages are used in trending markets. A sideways market is a market where the asset price stays in a fairly stable range.

Oscillators form as lines drawn below and above the price graph and their movement is set to a pre-defined range (Naved & Srivastava 2015). Oscillators are used to find out the overbought and oversold areas in the price chart of the asset value. In practice, if the price of an asset rises quickly the oscillator reacts to the change by reaching the level considered as overbought. This works to other direction as-well. If the value of the asset decreases rapidly, the oscillator reacts to it by reaching the level considered as oversold. These levels are the points where the trader needs to decide either to buy (at oversold level) or to sell (at overbought level) the asset. (Naved & Srivastava 2015)

The following chapters examines the theory behind two different types of oscillators used in cryptocurrency trading. These oscillators are the stochastic oscillator and the relative strength index (RSI).

4.3.1 Stochastic Oscillator

The stochastic oscillator is a classified as a momentum oscillator. It was invented by George C. Lane in the late 1950s. Lane discovered that in an up-trending stock, prices typically reach higher highs, with the daily closing price often clustering near the highest levels of recent periods (Hwa 2007). A similar pattern occurs with down-trending stocks, where the closing price tends to gather near the lowest levels of past periods (Hwa 2007). Based on these observations, Lane developed a mathematical formula for the stochastic oscillator, which is shown in figure 6.

$$\text{Stochastic} = \left(\frac{\text{Lowest share price during period} - \text{Last closing price}}{\text{Lowest share price during period} - \text{Highest closing price for the period}} \right) \times 100$$

Figure 6. The formula of the stochastic oscillator (Cohen & Cabiri 2015)

The oscillator oscillates in a fixed range in which the value of the indicator will decrease towards 0 in a down-trending market and towards 100 in an up-trending market making it possible to determine the oversold and overbought conditions (Hwa 2007). In addition, according to Hwa (2007), Lane established that a stochastic value of below 20 or above 80 may signal an imminent price trend reversal. When the oscillator crosses the trend line value of 20 from bottom to top it is a typical signal to buy and when it crosses the trend line value of 80 from top to bottom it is in turn a signal to sell (Cohen & Cabiri 2015).

4.3.2 Relative Strength Index (RSI)

According to Cohen & Cabiri (2015), the relative strength index (RSI) measures the rate of momentum that the asset in the market is gaining. In practice, the RSI indicator compares the upward movements in the closing price to its downward movements over a specified period, calculating the rate of increase or decrease (Cohen & Cabiri 2015; Navid & Srivastava 2015). The RSI was developed by Welles Wilder in 1978, and it is a widely used oscillator in technical analysis, known for its simplicity and ease of usage (Țăran-Moroșan 2011).

To calculate the RSI, the daily changes in the closing price are measured: the upward change (U) and the downward change (D), as shown in the formulas in figure 7.

$$U = \text{Close}_{\text{today}} - \text{Close}_{\text{yesterday}}\{D = 0\} \qquad D = \text{Close}_{\text{today}} - \text{Close}_{\text{yesterday}}\{U = 0\}$$

Figure 7. Upward change (U) and downward change (D) needed for the RSI (Cohen & Cabiri 2015)

The ratios between the upward change and the downward change are called the relative strength and this is a pivotal part of determining the indicator. The formula for determining the value of relative strength is represented by figure 8.

$$\text{RS} = \left|\frac{\text{EMA}[N] \text{ of } U}{\text{EMA}[N] \text{ of } D}\right|$$

Figure 8. The relative strength (Cohen & Cabiri 2015)

To clarify the symbols in figure 9, the EMA represents the exponential moving average of either the upward change (U) or the downward change (D). N represents the number of the days during which the price increases or decreases were examined. The base value is often 14 days, but the period can be increased in order to for example reduce the responsiveness of the indicator or vice versa (Cohen & Cabiri 2015). The relative is strength factor is then utilized to calculate the index which's formula is represented in figure 9.

$$\text{RSI} = \left|100 - \left(\frac{1}{1 + \text{RS}} \times 100\right)\right|$$

Figure 9. The formula of the Relative Strength Index (Cohen & Cabiri 2015)

The value of the relative strength index ranges between 0 to 100 similarly to the stochastic oscillator (Naved & Srivastava 2015). According to common market practice used by traders, RSI-value of over 70 indicates over-buying of an asset which means that the trader should consider selling the asset and in turn a value below 30 indicates over-selling which means that the trader should buy the asset (Cohen & Cabiri 2015). However, according to Ţăran-Moroşan (2011) in highly volatile markets such as the cryptocurrency market it is sometimes recommended to use values between 20 to 80 instead of 30 and 70 as buy- and sell-signals.

5 Backtesting Indicators on TradingView

This chapter will discuss how the different technical analysis methods performed in backtesting as-well as how they can be implemented in the cryptocurrency market on a basic level. A script for each indicator was inserted in the Pine Script editor on TradingView in order to get the results. The scripts for the indicator used will be included in the appendices 1 - 5. Bitcoin's price movements were used as the historical data for the procedure. The time frame for the test was set to start from 1.1.2022 and end during 1.1.2024 covering a two-year period in total as seen in figure 10. The course of Bitcoin was presented in euros. The reasoning for choosing a two-year period is due to it containing enough information in order to conduct a backtest with reasonably reliable results. The period also covers the most recent years in Bitcoin's price history which makes it possible for the results of the backtest to have relevance. However, it is possible to conduct the same backtest for a different period with a different length using the same methodology which is utilized in this research.

Figure 10. Bitcoin's price chart 1.1.2022 - 1.1.2024 (TradingView 2024)

5.1 Methodology

All the properties in TradingView were set as default while conducting the backtest in order for it to remain as simple as possible. This includes for example the starting capital of 1000000 and the commission of 0 %. It must be noted that these factors have an effect when it comes to the reliability of the results. In real trading these factors should be taken into account.

Furthermore, in the case of this study the main point is to implement the technical indicators in their basic form in order to see how they work and perform on a fundamental level without placing large emphasis on the quality of the trading strategy e.g. optimization. This helps to focus more on the implementation of the technical analysis in the cryptocurrency while including the ability analyse the performance of the indicators on a basic level. In real trading there are a lot more complex and advanced trading strategies which utilize the indicators discussed in this study more efficiently and take into account other different factors as-well that are outside the scope of this research.

For the simple moving average and the exponential moving average, a period length of 14 was used. For the Bollinger bands a length of 20 was used alongside with a band multiplier of two. The stochastic oscillator used a %K-Smoothing factor of 3 with a %D-Smoothing factor of the same value (3). The stochastic oscillator used 20 as the oversold level and 80 as the overbought level. The stochastic oscillator's length was set to 14. The relative strength index used the length of 14, 30 as the oversold level and 70 as the overbought level. The reason why these values were chosen is due them being classified as the most basic settings for the indicators chosen for this backtest. In another words TradingView sets for these values for the indicators automatically unless the user decides to change them manually. These values can of course be modified in order to get results that might differ radically compared to the results provided by this research. But due to the goal of keeping this research as simple as possible, the values set automatically by TradingView are being used since they still provide results that can be analysed and are comparable between each other.

5.2 Analysing the results

This chapter will provide the results of the back test alongside with an analysis on their performance. The data was downloaded from TradingView's strategy tester into an Excel-file and all the pivotal values were extracted as a separate table which was then screenshot and added to the thesis. The other values provided by the backtest will be in appendices 6 - 10. The analysis will focus on comparing each indicators' net profit, gross profit, gross loss, number of trades (total, winning and losing), largest losing & winning trades, average losing & winning trade and the win/loss-ratio of each indicator in order to rank their performance.

INDICATOR	Simple Moving Average (SMA)	Exponential Moving Average (EMA)	Bollinger Bands	Stochastic Oscillator	Relative Strength Index (RSI)
NET PROFIT	-32807.77	-28912.02	23625.71	842.86	11352.88
GROSS PROFIT	49226.62	50433.61	37258.22	3233.46	19185.98
GROSS LOSS	82034.39	79345.63	13632.51	2390.60	7833.1
NUMBER OF TRADES	109	125	29	40	4
NUMBER OF WINNING TRADES	25	25	20	16	3
NUMBER OF LOSING TRADES	84	100	9	24	1
LARGEST WINNING TRADE	7278.62	8548.42	6693.55	212.59	8063.28
LARGEST LOSING TRADE	3637.71	3637.71	8090.11	111.84	7833.1
AVERAGE WINNING TRADE	1969.06	2017.34	1862.91	202.09	6395.33
AVERAGE LOSING TRADE	976.6	793.46	1514.72	99.61	7833.1
WIN/LOSS RATIO	22.94	20	68.97	40	75

Figure 11. Values from different categories for the indicators gathered from the backtesting procedure

Figure 11 includes the values extracted from in different categories for the comparative analysis between the technical analysis indicators. These values were then ranked (Figure 12) in order to see how they performed compared to each other.

RANKING BY NET PROFIT

	INDICATOR	NET PROFIT
1)	Bollinger Bands	23625.71
2)	Relative Strength Index (RSI)	11352.88
3)	Stochastic Oscillator	842.86
4)	Exponential Moving Average (EMA)	-28912.02
5)	Simple Moving Average (SMA)	-32807.77

RANKING BY WIN-RATE

	INDICATOR	WIN/LOSS RATIO
1)	Relative Strength Index (RSI)	75
2)	Bollinger Bands	68.97
3)	Stochastic Oscillator	40
4)	Simple Moving Average (SMA)	22.94
5)	Exponential Moving Average (EMA)	20

RANKING BY GROSS PROFIT

	INDICATOR	GROSS PROFIT
1)	Exponential Moving Average (EMA)	50433.61
2)	Simple Moving Average (SMA)	49226.62
3)	Bollinger Bands	37258.22
4)	Relative Strength Index (RSI)	19185.98
5)	Stochastic Oscillator	3233.46

RANKING BY GROSS LOSS

	INDICATOR	GROSS LOSS
1)	Simple Moving Average (SMA)	82034.39
2)	Exponential Moving Average (EMA)	79345.63
3)	Bollinger Bands	13632.51
4)	Relative Strength Index (RSI)	7833.1
5)	Stochastic Oscillator	2390.60

RANKING BY AVERAGE WINNING TRADE

	INDICATOR	AVERAGE WINNING TRADE
2)	Relative Strength Index (RSI)	6395.33
3)	Exponential Moving Average (EMA)	2017.34
4)	Simple Moving Average (SMA)	1969.06
5)	Bollinger Bands	1862.91
1)	Stochastic Oscillator	202.19

RANKING BY AVERAGE LOSING TRADE

	INDICATOR	AVERAGE LOSING TRADE
1)	Relative Strength Index (RSI)	7833.1
2)	Bollinger Bands	1514.72
3)	Simple Moving Average (SMA)	976.6
4)	Exponential Moving Average (EMA)	793.46
5)	Stochastic Oscillator	99.61

RANKING BY LARGEST WINNING TRADE

	INDICATOR	LARGEST WINNING TRADE
1)	Exponential Moving Average (EMA)	8548.42
2)	Relative Strength Index (RSI)	8063.28
3)	Simple Moving Average (SMA)	7278.62
4)	Bollinger Bands	6693.55
5)	Stochastic Oscillator	212.59

RANKING BY LARGEST LOSING TRADE

	INDICATOR	LARGEST LOSING TRADE
1)	Bollinger Bands	8090.11
2)	Relative Strength Index (RSI)	7833.1
3)	Simple Moving Average (SMA)	3637.71
4)	Exponential Moving Average (EMA)	3637.71
5)	Stochastic Oscillator	111.84

RANKING BY NUMBER WINNING TRADES

	INDICATOR	NUMBER OF WINNING TRADES
1)	Simple Moving Average (SMA)	25
2)	Exponential Moving Average (EMA)	25
3)	Bollinger Bands	20
4)	Stochastic Oscillator	16
5)	Relative Strength Index (RSI)	3

RANKING BY NUMBER OF LOSING TRADES

	INDICATOR	NUMBER OF LOSING TRADES
1)	Exponential Moving Average (EMA)	100
2)	Simple Moving Average (SMA)	84
3)	Stochastic Oscillator	24
4)	Bollinger Bands	9
5)	Relative Strength Index (RSI)	1

RANKING BY NUMBER OF TRADES

	INDICATOR	NUMBER OF TRADES
1)	Exponential Moving Average (EMA)	125
2)	Simple Moving Average (SMA)	109
3)	Stochastic Oscillator	40
4)	Bollinger Bands	29
5)	Relative Strength Index (RSI)	4

Figure 12. Indicators values ranked by category

According to the ranking (Figure 12) the best performer out of the indicators by net profit during the testing period was Bollinger bands with the value of 23625,71 €. Relative strength index performed the second best while the moving averages (SMA & EMA) were the least profitable.

When it comes to the win-rate, the relative strength index was with the best with the ratio of 75 % followed by Bollinger bands with 68,97 % and stochastic oscillator with 40 %. The moving averages had the worst rating in this category as-well. The simple moving average had an atrocious win-rate of only 22,94 % while the exponential moving average performed even worse having a win-rate of 20 %.

However, it must be kept in mind that while the moving averages had a bad performance, they also had the most trades in total. The exponential moving had a total number of 125 trades, 100 of them being losing and 25 of them being winning. The simple moving average had the second largest number of trades, 109 out of which 25 were winning and 84 losing. Both of these indicators were not optimized for the maximum performance possible since this requires a lot of testing and experience. Due to this fact the indicators might give a lot of false signals due to the number of trades the back testing script forces them to commit to.

Because the indicators had more trading signals during the backtest this also explains why they had the largest gross profits and gross losses. Both of the moving averages traded the largest amount of Bitcoin during the test period out of all the other indicators.

The stochastic oscillator had a total number of 40 trades 16 of them resulting in a win and 24 in a loss. Bollinger bands traded 29 times winning 20 positions while losing only 9. The relative strength index had an extremely low number trades during the whole 2-year testing period. The relative strength index traded a total of 4 times winning positions 3 in total.

When it comes to the largest winning trades the exponential moving average had the best results having a winning trade of 8548,42 €. This was closely followed by the relative strength index with a winning trade of 8063,28 €. All of the winning trades were in the range of 6500 – 8500 € approximately except for the stochastic oscillator which had a very low largest winning trade of only 212,59 € relative to the other indicators.

In the category of the largest losing trades Bollinger bands ranked first followed by the relative strength index. The simple moving averages both had an equal score of 3637,71 € lost. The stochastic oscillator ranked the last with a largest losing trade of 111,84 €.

The relative strength had the highest average winning and losing trades. The stochastic oscillator had in turn the lowest.

The moving averages in general had the worst performance when it comes to the profitability. Both had a negative net profit meaning that during the time frame of the test from 1.1.2022 to 1.1.2024 using these indicators would have been costly for the trader.

5.3 Discussion

In summary, the indicators varied a lot in the results of the ranking. In order to determine which one of them was the best it would make sense to use the net profit and win-rate as the determining values. The reasoning for using net profit as the key determining value is due the fact that it describes how much money the indicator has generated during the research period. This is naturally the most interesting value for the trader to observe because one of the main goals in trading real-world assets is to generate the maximum amount of profit possible.

Win-rate is another good choice since it tells the trader how well the strategy or the indicator has performed during the research period. Usually a win-rate above 50 % means that the strategy or the indicator has managed take more winning trades compared to losing trades. In other words, the trader can assume that by utilizing this kind of a strategy they will gain more in the long run than lose. However, it must be noted that even high win-rate strategies can have losing streaks which makes them look bad in different periods depending on the timeframe of the data.

Figure 13 contains a short summary of the best performing indicator in each category that was chosen for further analysis in this research. According to figure 13, the best net-profit was obtained with the Bollinger bands indicator. The best win-rate in the other hand was obtained using the relative strength index The Bollinger bands and the RSI outperformed the stochastic oscillator and both moving averages in the categories of net profit and win-rate so in the framework of this study they should be considered as the best. When it comes to ranking the performance of the indicators based on pure profit generated it could be stated that the Bollinger bands had the best performance out of all the indicators included in this research.

SUMMARY OF BEST (WORST) IN CATEGORY

CATEGORY	INDICATOR
NETPROFIT	*Bollinger Bands*
WIN-RATE	*Relative Strength Index (RSI)*
GROSS PROFIT	*Exponential Moving Average (EMA)*
GROSS LOSS	*Simple Moving Average (SMA)*
AVERAGE WINNING TRADE	*Relative Strength Index (RSI)*
AVERAGE LOSING TRADE	*Relative Strength Index (RSI)*
LARGEST WINNING TRADE	*Exponential Moving Average (EMA)*
LARGEST LOSING TRADE	*Bollinger Bands*
NUMBER OF WINNING TRADES	*Simple Moving Average (SMA)*
NUMBER OF LOSING TRADES	*Exponential Moving Average (EMA)*
HIGHEST NUMBER OF TRADES	*Exponential Moving Average (EMA)*

Figure 13. Summary of the best indicator in each separate category

The research questions were the following:

Q1: How technical analysis can be utilized in cryptocurrency trading?

Q2: How have the common technical analysis indicators performed compared to each other in the Bitcoin cryptocurrency market?

According to the results of the research this thesis was able to provide an answer to both of these questions. First question was answered by implementing the technical indicators utilizing Pine Script in order to gather conclusive results based on the output of the scripts. The second question was answered based on the data presented by the different backtests conducted on each singular technical indicator. It could be said that this research has been a successful in determining concrete answers to the research problems.

When it comes to other categories further research could be conducted on for example gross profits and gross losses since they are being dominated by the moving averages. This could be for example due the fact that both of these indicators made more trades meaning that they have a larger sample size to conduct analysis on and for example point out singular large winning trades or streaks of successful trades in a row if they appear in the data.

It should be noted that the settings used for the indicators to gather these results may differ from the real-world trading conditions. Each indicator should be tested multiple times with different settings in order to maximize the profit generated by the strategy during the backtesting procedure. This is done because the trader wants to have multiple confirmations provided by multiple backtests on a

singular indicator in order to be certain of the profitability of a strategy. Therefore, when it comes to further research more backtests could be conducted on for example the Bollinger bands or the RSI-indicator in order to further optimize them and analyse the change in performance. But as stated before even though the indicator could be in theory optimized to near perfection, they can still fail if utilized in real trading due to the unpredictability of the market.

5.4 Limitations

The main limiting factor of this study is that it will focus only on Bitcoin only thus it cannot be generalized to the market as a whole because of the existence of multiple different cryptocurrency tokens and different types of blockchain technologies. Because of this, it is possible that there are differences in the behaviour of tokens which can be caused by multiple different factors.

Causes for the differences in behaviour can be for example different types of investor groups reacting differently when it comes to each unique token listed on the market. There is also the market structure which can create differences with the daily or monthly trading volume on each currency. There is also the fact that blockchains update on a different time basis. For example, Bitcoin updates its blockchain every ten minutes while on the other hand, Etherium is way faster and updates on a time scale between ten to fourteen seconds.

Another limiting factor is that the research is conducted on a set time scale. This research covers only a two-year period starting from 1.1.2022 and ending in 1.1.2024. This means that the profitability and performance of the indicators is limited to this time frame only and it is almost certain that the results will differ for example on a broader time scale.

It must be also noted that the results of this study are based on a backtest. The point of a backtest is to evaluate the effectiveness of a trading strategy i.e. it measures how a trading strategy has performed during the past market conditions. One reason for this approach is to discover the highest possible performance in theory, but it does not reflect real-world trading conditions. In live trading, risk management techniques like position sizing and margin controls are essential. Additionally, backtesting ignores transaction fees, slippage, and other trading costs, which can significantly impact the actual performance of a trading strategy.

6 Conclusions

This study was conducted in order to gain an overview of technical analysis and the indicators used in technical trading as-well as to provide a brief insight to what Bitcoin is and how its blockchain operates. The indicators chosen were the simple moving average (SMA), the exponential moving (EMA), Bollinger bands, stochastic oscillator, and the relative strength index (RSI) due to them being widely known popular indicators used in trading different types of assets, including cryptocurrencies.

Technical analysis and the indicators were discussed in detail and in theory in order to provide an overview on how they work in theory by using different academic literature sources associated with them. Technical analysis and its indicators are used to research different financial markets in order to forecast future price movements based on historical data. Compared to fundamental analysis, which evaluates an assets intrinsic value by analysing different financial and economic factors, technical analysis focuses more on finding patterns from the data such as trends, and price changes. Technical analysis has a vast set of different indicators and not all of them could be included in this study.

The indicators that were implemented in practice by conducting a backtest based on Bitcoin's historical data on a platform called TradingView by utilizing TradingViews own Strategy Tester which utilizes Pine Script as its programming language. The reason why backtest was chosen as the method was because it is a fundamental part of technical analysis which utilizes the indicators or a strategy consisting of multiple different indicators in order to examine their performance in the past. It it also a safer method compared to investing in Bitcoin without any research. The time frame for the test was set to start at 1.1.2022 and to end at 1.1.2024. Pine Script ran simulations in TradingView's Strategy Tester in order determine how profitable the chosen indicators were during the set time frame.

The results of the backtest varied between the indicators. Based on profitability as the key determining factor of performance the Bollinger bands and the relative strength index had the best performance out of the five chosen indicators when it came to netprofit and win-rate. In turn the simple moving average and the exponential moving performed worst in the terms of profitability achieving a negative net profit. However, it must be noted that the indicators used were very simple and they were also not optimized for maximum performance due to the scope of this research.

The results of this study can be utilized in many ways. The main purpose of this study was to introduce technical analysis and the theory behind it which is useful knowledge for anyone interested in trading

assets such as cryptocurrencies. The script for the indicators can be utilized for example backtesting other cryptocurrencies on TradingView's Strategy Tester or order to get a basic understanding how to implement indicators to Pine Script.

Technical analysis is a broad and expansive field, and this study addresses only a small portion of it on a very basic level. Future research could for example explore other indicators used in technical analysis or enhance the indicators used for this research here through various means. For example, the script for the indicators could be optimized better so that the backtest will give improved results compared to those of this study. Also, the indicators can be combined and modified further in order generate an actual high performance trading strategy for cryptocurrency trading. Many of the parameters used to conduct the backtest can be changed in order to research for instance a completely different time frame resulting in new results.

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APPENDICES

Appendix 1. Script for Backtesting the Simple Moving Average

```
//@version=6
strategy("SMA Backtest", overlay=true)

// Input for SMA period
smaLength = input.int(14, title="SMA Length", minval=1)

// Define the start and end dates for the time frame
startDate = timestamp(2022, 01, 01, 00, 00)  // January 1, 2022
endDate = timestamp(2024, 01, 01, 00, 00)    // January 1, 2024

// Check if the current bar is within the defined time frame
inTimeFrame = (time >= startDate and time <= endDate)

// Calculate the Simple Moving Average
smaValue = ta.sma(close, smaLength)

// Define trading conditions
longCondition = ta.crossover(close, smaValue) and inTimeFrame
shortCondition = ta.crossunder(close, smaValue) and inTimeFrame

// Enter long position when price crosses above the SMA
if (longCondition)
    strategy.entry("Long", strategy.long)

// Enter short position when price crosses below the SMA
if (shortCondition)
    strategy.entry("Short", strategy.short)

// Close long position when price crosses below the SMA
if (ta.crossunder(close, smaValue) and strategy.position_size > 0)
    strategy.close("Long")
```

```
// Close short position when price crosses above the SMA
if (ta.crossover(close, smaValue) and strategy.position_size < 0)
    strategy.close("Short")

// Plot the Simple Moving Average
plot(smaValue, color=color.blue, title="SMA")
```

Appendix 2. Script for Backtesting the Exponential Moving Average

```
//@version=6
strategy("EMA Backtest" , overlay=true)

// Input for EMA period
emaLength = input.int(14, title="EMA Length", minval=1)

// Define the start and end dates for the time frame
startDate = timestamp(2022, 01, 01, 00, 00)  // January 1, 2022
endDate = timestamp(2024, 01, 01, 00, 00)    // January 1, 2024

// Check if the current bar is within the defined time frame
inTimeFrame = (time >= startDate and time <= endDate)

// Calculate the Exponential Moving Average
emaValue = ta.ema(close, emaLength)

// Define trading conditions
longCondition = ta.crossover(close, emaValue) and inTimeFrame
shortCondition = ta.crossunder(close, emaValue) and inTimeFrame

// Enter long position when price crosses above the EMA
if (longCondition)
    strategy.entry("Long", strategy.long)
```

```
// Enter short position when price crosses below the EMA
if (shortCondition)
    strategy.entry("Short", strategy.short)

// Close long position when price crosses below the EMA
if (ta.crossunder(close, emaValue) and strategy.position_size > 0)
    strategy.close("Long")

// Close short position when price crosses above the EMA
if (ta.crossover(close, emaValue) and strategy.position_size < 0)
    strategy.close("Short")

// Plot the Exponential Moving Average
plot(emaValue, color=color.blue, title="EMA")
```

Appendix 3. Script for Backtesting Bollinger Bands

```
//@version=6
strategy("Bollinger Bands Backtest", overlay=true)

// Input for Bollinger Bands
bbLength = input.int(20, title="Bollinger Bands Length", minval=1)
bbMult = input.float(2.0, title="Bollinger Bands Multiplier", minval=0.1)

// Define the start and end dates for the time frame
startDate = timestamp(2022, 01, 01, 00, 00)  // January 1, 2022
endDate = timestamp(2024, 01, 01, 00, 00)    // January 1, 2024

// Check if the current bar is within the defined time frame
inTimeFrame = (time >= startDate and time <= endDate)
```

```
// Calculate the Bollinger Bands
basis = ta.sma(close, bbLength)
dev = bbMult * ta.stdev(close, bbLength)
upper = basis + dev
lower = basis - dev

// Define trading conditions
longCondition = ta.crossover(close, lower) and inTimeFrame
shortCondition = ta.crossunder(close, upper) and inTimeFrame

// Enter long position when price crosses above the lower Bollinger Band
if (longCondition)
    strategy.entry("Long", strategy.long)

// Enter short position when price crosses below the upper Bollinger Band
if (shortCondition)
    strategy.entry("Short", strategy.short)

// Close long position when price crosses below the basis (SMA)
if (ta.crossunder(close, basis) and strategy.position_size > 0)
    strategy.close("Long")

// Close short position when price crosses above the basis (SMA)
if (ta.crossover(close, basis) and strategy.position_size < 0)
    strategy.close("Short")

// Plot the Bollinger Bands
plot(upper, color=color.green, title="Upper Bollinger Band", linewidth=2)
plot(lower, color=color.red, title="Lower Bollinger Band", linewidth=2)
plot(basis, color=color.blue, title="Middle Bollinger Band", linewidth=1)
```

Appendix 4. Script for Backtesting Stochastic Oscillator

```
//@version=6
strategy("Stochastic Oscillator Backtest", overlay=true)

// Input for Stochastic Oscillator
stochasticLength = input.int(14, title="Stochastic Length", minval=1)
smoothK = input.int(3, title="%K Smoothing", minval=1)
smoothD = input.int(3, title="%D Smoothing", minval=1)
oversoldLevel = input.int(20, title="Oversold Level", minval=1, maxval=50)
overboughtLevel = input.int(80, title="Overbought Level", minval=50, maxval=100)

// Input for time frame
startDate = timestamp(2022, 01, 01, 00, 00)  // January 1, 2022
endDate = timestamp(2024, 01, 01, 00, 00)    // January 1, 2024

// Check if the current bar is within the defined time frame
inTimeFrame = (time >= startDate and time <= endDate)

// Calculate Stochastic Oscillator %K
k = ta.stoch(close, high, low, stochasticLength)
smoothKValue = ta.sma(k, smoothK)

// Calculate Stochastic Oscillator %D (Signal line)
d = ta.sma(smoothKValue, smoothD)

// Define conditions for entering a long position
longCondition = ta.crossover(smoothKValue, d) and smoothKValue < oversoldLevel
and inTimeFrame

// Define conditions for entering a short position
shortCondition = ta.crossunder(smoothKValue, d) and smoothKValue >
overboughtLevel and inTimeFrame

// Define Stop Loss and Take Profit (in price units)
stopLoss = input.float(100, title="Stop Loss (in price)", minval=0)
takeProfit = input.float(200, title="Take Profit (in price)", minval=0)
```

```
// Enter long position when the Stochastic Oscillator %K crosses above the %D
if (longCondition)
    strategy.entry("Long", strategy.long)

// Enter short position when the Stochastic Oscillator %K crosses below the %D
if (shortCondition)
    strategy.entry("Short", strategy.short)

// Exit conditions with Stop Loss and Take Profit using strategy.exit
strategy.exit("Exit Long", from_entry="Long", stop=close - stopLoss, limit=close
+ takeProfit)
strategy.exit("Exit Short", from_entry="Short", stop=close + stopLoss,
limit=close - takeProfit)

// Plot the Stochastic Oscillator %K and %D on the chart
plot(smoothKValue, color=color.green, title="%K", linewidth=2)
plot(d, color=color.red, title="%D", linewidth=2)
hline(oversoldLevel, "Oversold Level", color=color.green,
linestyle=hline.style_dotted)
hline(overboughtLevel, "Overbought Level", color=color.red,
linestyle=hline.style_dotted)
```

Appendix 5. Script for Backtesting the Relative Strength Index

```
//@version=6
strategy("RSI Backtest", overlay=true)

// Input for RSI length and overbought/oversold levels
rsiLength = input.int(14, title="RSI Length", minval=1)
oversoldLevel = input.int(30, title="Oversold Level", minval=1, maxval=50)
overboughtLevel = input.int(70, title="Overbought Level", minval=50, maxval=100)

// Time frame for backtest
startDate = timestamp(2022, 01, 01, 00, 00)  // January 1, 2022
```

```
endDate = timestamp(2024, 01, 01, 00, 00)    // January 1, 2024

// Check if the current bar is within the defined time frame
inTimeFrame = (time >= startDate and time <= endDate)

// Calculate RSI
rsiValue = ta.rsi(close, rsiLength)

// Define long and short conditions
longCondition = ta.crossover(rsiValue, oversoldLevel) and inTimeFrame
shortCondition = ta.crossunder(rsiValue, overboughtLevel) and inTimeFrame

// Enter long position when RSI crosses above the oversold level
if (longCondition)
    strategy.entry("Long", strategy.long)

// Exit long position when RSI crosses below the overbought level
if (shortCondition)
    strategy.close("Long")

// Plot RSI and overbought/oversold levels
plot(rsiValue, color=color.blue, title="RSI", linewidth=2)
hline(oversoldLevel, "Oversold Level", color=color.green,
linestyle=hline.style_dotted)
hline(overboughtLevel, "Overbought Level", color=color.red,
linestyle=hline.style_dotted)
```

Appendix 6. Summary of all the values from the Simple Moving Average Strategy

SUMMARY (SMA)	All EUR	All %	Long EUR	Long %	Short EUR	Short %
Net Profit	-32807.77	-3.28	-13980.67	-1.4	-18827.1	-1.88
Gross Profit	49226.62	4.92	25847.05	2.58	23379.57	2.34
Gross Loss	82034.39	8.2	39827.72	3.98	42206.67	4.22
Max Run-up	12677.18	1.28				
Max Drawdown	34608.02	3.46				
Buy & Hold Return	1742623.69	174.26				
Sharpe Ratio	-0.909					
Sortino Ratio	-0.688					
Profit Factor	0.6		0.649		0.554	
Max Contracts Held	1		1		1	
Open PL	0	0				
Commission Paid	0		0		0	
Total Closed Trades	109		55		54	
Total Open Trades	0		0		0	
Number Winning Trades	25		13		12	
Number Losing Trades	84		42		42	
Percent Profitable	22.94		23.64		22.22	
Avg Trade	-300.99	-0.8	-254.19	-0.44	-348.65	-1.17
Avg Winning Trade	1969.06	7.88	1988.23	8.44	1948.3	7.28
Avg Losing Trade	976.6	3.39	948.28	3.19	1004.92	3.59
Ratio Avg Win / Avg Loss	2.016		2.097		1.939	
Largest Winning Trade	7278.62	34.61	5689.89	34.61	7278.62	26.34
Largest Losing Trade	3637.71	12.02	3165.63	8.22	3637.71	12.02
Avg # Bars in Trades	8		8		7	
Avg # Bars in Winning Trades	17		17		16	
Avg # Bars in Losing Trades	5		5		5	
Margin Calls	0		0		0	

Appendix 7. Summary of all the values from the Exponential Moving Average Strategy

SUMMARY (EMA)	All EUR	All %	Long EUR	Long %	Short EUR	Short %
Net Profit	-28912.02	-2.89	-12060	-1.21	-16852	-1.69
Gross Profit	50433.61	5.04	26654.23	2.67	23779.38	2.38
Gross Loss	79345.63	7.93	38714.21	3.87	40631.42	4.06
Max Run-up	15945.88	1.61				
Max Drawdown	30712.27	3.07				
Buy & Hold Return	1738755.4	173.9				
Sharpe Ratio	-0.769					
Sortino Ratio	-0.633					
Profit Factor	0.636		0.688		0.585	
Max Contracts Held	1		1		1	
Open PL	0	0				
Commission Paid	0		0		0	
Total Closed Trades	125		63		62	
Total Open Trades	0		0		0	
Number Winning Trades	25		13		12	
Number Losing Trades	100		50		50	
Percent Profitable	20		20.63		19.35	
Avg Trade	-231.3	-0.54	-191.43	-0.23	-271.81	-0.85
Avg Winning Trade	2017.34	8.01	2050.33	8.61	1981.61	7.36
Avg Losing Trade	793.46	2.68	774.28	2.53	812.63	2.82
Ratio Avg Win / Avg Loss	2.542		2.648		2.439	
Largest Winning Trade	8548.42	30.33	6876.41	28.49	8548.42	30.33
Largest Losing Trade	3637.71	12.02	2692.57	7.21	3637.71	12.02
Avg # Bars in Trades	7		7		7	
Avg # Bars in Winning Trades	18		18		17	
Avg # Bars in Losing Trades	4		4		4	
Margin Calls	0		0		0	

Appendix 8. Summary of all the values from the Bollinger Bands Strategy

SUMMARY (BOLLINGER)	All EUR	All %	Long EUR	Long %	Short EUR	Short %
Net Profit	23625.71	2.36	19117.12	1.91	4508.59	0.45
Gross Profit	37258.22	3.73	21999.61	2.2	15258.61	1.53
Gross Loss	13632.51	1.36	2882.49	0.29	10750.02	1.08
Max Run-up	33252.5	3.22				
Max Drawdown	11106.39	1.08				
Buy & Hold Return	1560897.98	156.09				
Sharpe Ratio	-0.239					
Sortino Ratio	-0.27					
Profit Factor	2.733		7.632		1.419	
Max Contracts Held	1		1		1	
Open PL	0	0				
Commission Paid	0		0		0	
Total Closed Trades	29		13		16	
Total Open Trades	0		0		0	
Number Winning Trades	20		9		11	
Number Losing Trades	9		4		5	
Percent Profitable	68.97		69.23		68.75	
Avg Trade	814.68	3.89	1470.55	7.37	281.79	1.06
Avg Winning Trade	1862.91	8.3	2444.4	11.94	1387.15	5.33
Avg Losing Trade	1514.72	5.92	720.62	2.91	2150	8.33
Ratio Avg Win / Avg Loss	1.23		3.392		0.645	
Largest Winning Trade	6693.55	34.78	6693.55	34.78	4168.7	20.48
Largest Losing Trade	8090.11	30.11	1227.6	5.73	8090.11	30.11
Avg # Bars in Trades	17		18		16	
Avg # Bars in Winning Trades	16		20		12	
Avg # Bars in Losing Trades	20		16		23	
Margin Calls	0		0		0	

Appendix 9. Summary of all the values from the Stochastic Oscillator Strategy

SUMMARY (STOCHASTIC OSC.)	All EUR	All %	Long EUR	Long %	Short EUR	Short %
Net Profit	842.86	0.08	229.08	0.02	613.78	0.06
Gross Profit	3233.46	0.32	1412.62	0.14	1820.84	0.18
Gross Loss	2390.6	0.24	1183.54	0.12	1207.06	0.12
Max Run-up	1307.03	0.13				
Max Drawdown	623.68	0.06				
Buy & Hold Return	1549114.5	154.9				
Sharpe Ratio	-10.61					
Sortino Ratio	-0.996					
Profit Factor	1.353		1.194		1.508	
Max Contracts Held	1		1		1	
Open PL	0	0				
Commission Paid	0		0		0	
Total Closed Trades	40		19		21	
Total Open Trades	0		0		0	
Number Winning Trades	16		7		9	
Number Losing Trades	24		12		12	
Percent Profitable	40		36.84		42.86	
Avg Trade	21.07	0.06	12.06	0	29.23	0.11
Avg Winning Trade	202.09	0.79	201.8	0.8	202.32	0.77
Avg Losing Trade	99.61	0.43	98.63	0.47	100.59	0.39
Ratio Avg Win / Avg Loss	2.029		2.046		2.011	
Largest Winning Trade	212.59	1.07	211	1.07	212.59	1.06
Largest Losing Trade	111.84	0.63	105.33	0.63	111.84	0.58
Avg # Bars in Trades	1		1		1	
Avg # Bars in Winning Trades	1		1		1	
Avg # Bars in Losing Trades	1		1		1	
Margin Calls	0		0		0	

Appendix 10. Summary of all the values from the Relative Strength Index Strategy

SUMMARY (RSI)	All EUR	All %	Long EUR	Long %	Short EUR	Short %
Net Profit	11352.88	1.14	11352.9	1.14	0	0
Gross Profit	19185.98	1.92	19186	1.92	0	0
Gross Loss	7833.1	0.78	7833.1	0.78	0	0
Max Run-up	15509.24	1.53				
Max Drawdown	13766.42	1.37				
Buy & Hold Return	1550152	155				
Sharpe Ratio	-0.4					
Sortino Ratio	-0.456					
Profit Factor	2.449		2.449		N/A	
Max Contracts Held	1		1		0	
Open PL	0	0				
Commission Paid	0		0		0	
Total Closed Trades	4		4		0	
Total Open Trades	0		0		0	
Number Winning Trades	3		3		0	
Number Losing Trades	1		1		0	
Percent Profitable	75		75		N/A	
Avg Trade	2838.22	12.69	2838.22	12.69	N/A	N/A
Avg Winning Trade	6395.33	25.97	6395.33	25.97	N/A	N/A
Avg Losing Trade	7833.1	27.13	7833.1	27.13	N/A	N/A
Ratio Avg Win / Avg Loss	0.816		0.816		N/A	
Largest Winning Trade	8063.28	33.13	8063.28	33.13	N/A	N/A
Largest Losing Trade	7833.1	27.13	7833.1	27.13	N/A	N/A
Avg # Bars in Trades	107		107		0	
Avg # Bars in Winning Trades	55		55		0	
Avg # Bars in Losing Trades	262		262		0	
Margin Calls	0		0		0	

Appendix 11. Properties for the Simple Moving Average

PROPERTIES (SMA)	
Title	Value
Trading range	2022-01-30 â€" 2024-12-08
Backtesting range	2015-04-23 â€" 2024-12-08
Symbol	COINBASE:BTCEUR
Timeframe	1 day
Chart type	Line
Point Value	1
Currency	EUR
Tick Size	0.01
Precision	Default
SMA Length	14
Initial capital	1000000 EUR
Order size	1 Contracts
Pyramiding	1 orders
Commission	0%
Slippage	0 ticks
Verify price for limit orders	0 ticks
Margin for long positions	100%
Margin for short positions	100%
Recalculate After order is filled	On
Recalculate On every tick	On
Recalculate On bar close	On
Backtesting precision. Use bar magnifier	On

Appendix 12. Properties for the Exponential Moving Average

PROPERTIES (EMA)	
Title	Value
Trading range	2022-02-02 â€” 2024-12-08
Backtesting range	2015-04-23 â€” 2024-12-08
Symbol	COINBASE:BTCEUR
Timeframe	1 day
Chart type	Line
Point Value	1
Currency	EUR
Tick Size	0.01
Precision	Default
EMA Length	14
Initial capital	1000000 EUR
Order size	1 Contracts
Pyramiding	1 orders
Commission	0%
Slippage	0 ticks
Verify price for limit orders	0 ticks
Margin for long positions	100%
Margin for short positions	100%
Recalculate After order is filled	On
Recalculate On every tick	On
Recalculate On bar close	On
Backtesting precision. Use bar magnifier	On

Appendix 13. Properties for the Bollinger bands

PROPERTIES (BOLLINGER)	
Title	Value
Trading range	2022-01-09 â€" 2024-12-08
Backtesting range	2015-04-23 â€" 2024-12-08
Symbol	COINBASE:BTCEUR
Timeframe	1 day
Chart type	Line
Point Value	1
Currency	EUR
Tick Size	0.01
Precision	Default
Bollinger Bands Length	20
Bollinger Bands Multiplier	2
Initial capital	1000000 EUR
Order size	1 Contracts
Pyramiding	1 orders
Commission	0%
Slippage	0 ticks
Verify price for limit orders	0 ticks
Margin for long positions	100%
Margin for short positions	100%
Recalculate After order is filled	On
Recalculate On every tick	On
Recalculate On bar close	On
Backtesting precision. Use bar magnifier	On

Appendix 14. Properties for the Stochastic oscillator

PROPERTIES (STOCHASTIC OSC.)

Title	Value
Trading range	2022-01-10 â€" 2024-12-08
Backtesting range	2015-04-23 â€" 2024-12-08
Symbol	COINBASE:BTCEUR
Timeframe	1 day
Chart type	Line
Point Value	1
Currency	EUR
Tick Size	0.01
Precision	Default
Stochastic Length	14
%K Smoothing	3
%D Smoothing	3
Oversold Level	20
Overbought Level	80
Stop Loss (in price)	100
Take Profit (in price)	200
Initial capital	1000000 EUR
Order size	1 Contracts
Pyramiding	1 orders
Commission	0%
Slippage	0 ticks
Verify price for limit orders	0 ticks
Margin for long positions	100%
Margin for short positions	100%
Recalculate After order is filled	On
Recalculate On every tick	On
Recalculate On bar close	On
Backtesting precision. Use bar magnifier	On

Appendix 15. Properties for the Relative Strength Index

PROPERTIES (RSI)	
Title	Value
Trading range	2022-01-10 â€" 2024-12-08
Backtesting range	2015-04-23 â€" 2024-12-08
Symbol	COINBASE:BTCEUR
Timeframe	1 day
Chart type	Line
Point Value	1
Currency	EUR
Tick Size	0.01
Precision	Default
RSI Length	14
Oversold Level	30
Overbought Level	70
Initial capital	1000000 EUR
Order size	1 Contracts
Pyramiding	1 orders
Commission	0%
Slippage	0 ticks
Verify price for limit orders	0 ticks
Margin for long positions	100%
Margin for short positions	100%
Recalculate After order is filled	On
Recalculate On every tick	On
Recalculate On bar close	On
Backtesting precision. Use bar magnifier	On

Appendix 16. List of trades (Simple Moving Average)

LIST OF TRADES (SMA)

Trade #	Type	Signal	Date/Time	Price EUR	Contracts	Profit EUR	Profit %	Cum. Profit EUR	Cum. Profit %	Run-up EUR	Run-up %	Drawdown EUR	Drawdown %
109	Exit Long	Close entry(s	13/01/2024	39150.25	1	-941.14	-2.35	-32807.77	-0.1	4796.6	11.96	2741.39	6.84
109	Entry Long	Long	02/01/2024	40091.39	1	-941.14	-2.35	-32807.77	-0.1	4796.6	11.96	2741.39	6.84
108	Exit Short	Long	02/01/2024	40091.39	1	-1581.66	-4.11	-31866.63	-0.16	965.42	2.51	1590.27	4.13
108	Entry Short	Short	29/12/2023	38509.73	1	-1581.66	-4.11	-31866.63	-0.16	965.42	2.51	1590.27	4.13
107	Exit Long	Short	29/12/2023	38509.73	1	-646.82	-1.65	-30284.97	-0.07	293.21	0.75	998.14	2.55
107	Entry Long	Long	28/12/2023	39156.55	1	-646.82	-1.65	-30284.97	-0.07	293.21	0.75	998.14	2.55
106	Exit Short	Long	28/12/2023	39156.55	1	-623.14	-1.62	-29638.15	-0.06	333.41	0.87	852.59	2.21
106	Entry Short	Short	27/12/2023	38533.41	1	-623.14	-1.62	-29638.15	-0.06	333.41	0.87	852.59	2.21
105	Exit Long	Short	27/12/2023	38533.41	1	-1357.83	-3.4	-29015.01	-0.14	498.76	1.25	2135.85	5.35
105	Entry Long	Long	21/12/2023	39891.24	1	-1357.83	-3.4	-29015.01	-0.14	498.76	1.25	2135.85	5.35
104	Exit Short	Long	21/12/2023	39891.24	1	-1368.44	-3.55	-27657.18	-0.14	1381.55	3.59	1905.3	4.95
104	Entry Short	Short	16/12/2023	38522.8	1	-1368.44	-3.55	-27657.18	-0.14	1381.55	3.59	1905.3	4.95
103	Exit Long	Short	16/12/2023	38522.8	1	4147.42	12.07	-26288.74	0.43	7207.27	20.97	772.38	2.25
103	Entry Long	Long	23/11/2023	34375.38	1	4147.42	12.07	-26288.74	0.43	7207.27	20.97	772.38	2.25
102	Exit Short	Long	23/11/2023	34375.38	1	-1603.87	-4.89	-30436.16	-0.17	94.02	0.29	2012.77	6.14
102	Entry Short	Short	22/11/2023	32771.51	1	-1603.87	-4.89	-30436.16	-0.17	94.02	0.29	2012.77	6.14
101	Exit Long	Short	22/11/2023	32771.51	1	-1549.25	-4.51	-28832.29	-0.16	222.24	0.65	1559.42	4.54
101	Entry Long	Long	20/11/2023	34320.76	1	-1549.25	-4.51	-28832.29	-0.16	222.24	0.65	1559.42	4.54
100	Exit Short	Long	20/11/2023	34320.76	1	-982.94	-2.95	-27283.04	-0.1	307.21	0.92	1085.43	3.26
100	Entry Short	Short	17/11/2023	33337.82	1	-982.94	-2.95	-27283.04	-0.1	307.21	0.92	1085.43	3.26
99	Exit Long	Short	17/11/2023	33337.82	1	-1618.99	-4.63	-26300.1	-0.17	33.27	0.1	2246.81	6.43
99	Entry Long	Long	16/11/2023	34956.81	1	-1618.99	-4.63	-26300.1	-0.17	33.27	0.1	2246.81	6.43
98	Exit Short	Long	16/11/2023	34956.81	1	-2250.52	-6.88	-24681.11	-0.23	172.43	0.53	2315.24	7.08
98	Entry Short	Short	15/11/2023	32706.29	1	-2250.52	-6.88	-24681.11	-0.23	172.43	0.53	2315.24	7.08
97	Exit Long	Short	15/11/2023	32706.29	1	5689.89	21.06	-22430.59	0.59	8457.88	31.31	403.26	1.49
97	Entry Long	Long	17/10/2023	27016.4	1	5689.89	21.06	-22430.59	0.59	8457.88	31.31	403.26	1.49
96	Exit Short	Long	17/10/2023	27016.4	1	-1192.83	-4.62	-28120.48	-0.12	754.64	2.92	2736.42	10.6
96	Entry Short	Short	11/10/2023	25823.57	1	-1192.83	-4.62	-28120.48	-0.12	754.64	2.92	2736.42	10.6
95	Exit Long	Short	11/10/2023	25823.57	1	745.71	2.97	-26927.65	0.08	2118.41	8.45	41.4	0.17
95	Entry Long	Long	28/09/2023	25077.86	1	745.71	2.97	-26927.65	0.08	2118.41	8.45	41.4	0.17
94	Exit Short	Long	28/09/2023	25077.86	1	-423.53	-1.72	-27673.36	-0.04	239.33	0.97	793.36	3.22
94	Entry Short	Short	25/09/2023	24654.33	1	-423.53	-1.72	-27673.36	-0.04	239.33	0.97	793.36	3.22
93	Exit Long	Short	25/09/2023	24654.33	1	213.36	0.87	-27249.83	0.02	1291.29	5.28	95.42	0.39
93	Entry Long	Long	14/09/2023	24440.97	1	213.36	0.87	-27249.83	0.02	1291.29	5.28	95.42	0.39
92	Exit Short	Long	14/09/2023	24440.97	1	-221.86	-0.92	-27463.19	-0.02	1044.11	4.31	512.01	2.11
92	Entry Short	Short	09/09/2023	24219.11	1	-221.86	-0.92	-27463.19	-0.02	1044.11	4.31	512.01	2.11
91	Exit Long	Short	09/09/2023	24219.11	1	-334.94	-1.36	-27241.33	-0.03	137.45	0.56	578.68	2.36
91	Entry Long	Long	08/09/2023	24554.05	1	-334.94	-1.36	-27241.33	-0.03	137.45	0.56	578.68	2.36
90	Exit Short	Long	08/09/2023	24554.05	1	-644.33	-2.69	-26906.39	-0.07	440.72	1.84	803.61	3.36
90	Entry Short	Short	01/09/2023	23909.72	1	-644.33	-2.69	-26906.39	-0.07	440.72	1.84	803.61	3.36
89	Exit Long	Short	01/09/2023	23909.72	1	-1574.47	-6.18	-26262.06	-0.16	45.37	0.18	1811.86	7.11
89	Entry Long	Long	30/08/2023	25484.19	1	-1574.47	-6.18	-26262.06	-0.16	45.37	0.18	1811.86	7.11
88	Exit Short	Long	30/08/2023	25484.19	1	902.47	3.42	-24687.59	0.09	3273.66	12.41	55.6	0.21
88	Entry Short	Short	17/08/2023	26386.66	1	902.47	3.42	-24687.59	0.09	3273.66	12.41	55.6	0.21
87	Exit Long	Short	17/08/2023	26386.66	1	-760.93	-2.8	-25590.06	-0.08	270.18	1	772.79	2.85
87	Entry Long	Long	09/08/2023	27147.59	1	-760.93	-2.8	-25590.06	-0.08	270.18	1	772.79	2.85
86	Exit Short	Long	09/08/2023	27147.59	1	-496.44	-1.86	-24829.13	-0.05	571.15	2.14	918.85	3.45
86	Entry Short	Short	03/08/2023	26651.15	1	-496.44	-1.86	-24829.13	-0.05	571.15	2.14	918.85	3.45
85	Exit Long	Short	03/08/2023	26651.15	1	-324.1	-1.2	-24332.69	-0.03	319.12	1.18	565.7	2.1
85	Entry Long	Long	02/08/2023	26975.25	1	-324.1	-1.2	-24332.69	-0.03	319.12	1.18	565.7	2.1
84	Exit Short	Long	02/08/2023	26975.25	1	71.27	0.26	-24008.59	0.01	988.51	3.65	245.57	0.91
84	Entry Short	Short	15/07/2023	27046.52	1	71.27	0.26	-24008.59	0.01	988.51	3.65	245.57	0.91
83	Exit Long	Short	15/07/2023	27046.52	1	-980.17	-3.5	-24079.86	-0.1	127.15	0.45	1386.69	4.95
83	Entry Long	Long	14/07/2023	28026.69	1	-980.17	-3.5	-24079.86	-0.1	127.15	0.45	1386.69	4.95
82	Exit Short	Long	14/07/2023	28026.69	1	-562.34	-2.05	-23099.69	-0.06	332.51	1.21	884.59	3.22

82	Entry Short	Short	07/07/2023	27464.35	1	-562.34	-2.05	-23099.69	-0.06	332.51	1.21	884.59	3.22
81	Exit Long	Short	07/07/2023	27464.35	1	2891.6	11.77	-22537.35	0.3	4425.48	18.01	156.25	0.64
81	Entry Long	Long	20/06/2023	24572.75	1	2891.6	11.77	-22537.35	0.3	4425.48	18.01	156.25	0.64
80	Exit Short	Long	20/06/2023	24572.75	1	18.5	0.08	-25428.95	0	1809	7.36	286.37	1.16
80	Entry Short	Short	08/06/2023	24591.25	1	18.5	0.08	-25428.95	0	1809	7.36	286.37	1.16
79	Exit Long	Short	08/06/2023	24591.25	1	-862.36	-3.39	-25447.45	-0.09	135.54	0.53	1033.61	4.06
79	Entry Long	Long	07/06/2023	25453.61	1	-862.36	-3.39	-25447.45	-0.09	135.54	0.53	1033.61	4.06
78	Exit Short	Long	07/06/2023	25453.61	1	-1421.96	-5.92	-24585.09	-0.15	278.43	1.16	1538.32	6.4
78	Entry Short	Short	06/06/2023	24031.65	1	-1421.96	-5.92	-24585.09	-0.15	278.43	1.16	1538.32	6.4
77	Exit Long	Short	06/06/2023	24031.65	1	-1420.35	-5.58	-23163.13	-0.15	193.92	0.76	1728.79	6.79
77	Entry Long	Long	03/06/2023	25452	1	-1420.35	-5.58	-23163.13	-0.15	193.92	0.76	1728.79	6.79
76	Exit Short	Long	03/06/2023	25452	1	-516.75	-2.07	-21742.78	-0.05	287.33	1.15	566.75	2.27
76	Entry Short	Short	02/06/2023	24935.25	1	-516.75	-2.07	-21742.78	-0.05	287.33	1.15	566.75	2.27
75	Exit Long	Short	02/06/2023	24935.25	1	15.99	0.06	-21226.03	0	1622.76	6.51	156.26	0.63
75	Entry Long	Long	27/05/2023	24919.26	1	15.99	0.06	-21226.03	0	1622.76	6.51	156.26	0.63
74	Exit Short	Long	27/05/2023	24919.26	1	-444.48	-1.82	-21242.02	-0.05	389.89	1.59	670.95	2.74
74	Entry Short	Short	25/05/2023	24474.78	1	-444.48	-1.82	-21242.02	-0.05	389.89	1.59	670.95	2.74
73	Exit Long	Short	25/05/2023	24474.78	1	-797.69	-3.16	-20797.54	-0.08	7.04	0.03	1048.37	4.15
73	Entry Long	Long	24/05/2023	25272.47	1	-797.69	-3.16	-20797.54	-0.08	7.04	0.03	1048.37	4.15
72	Exit Short	Long	24/05/2023	25272.47	1	-553.26	-2.24	-19999.85	-0.06	197.71	0.8	707.6	2.86
72	Entry Short	Short	22/05/2023	24719.21	1	-553.26	-2.24	-19999.85	-0.06	197.71	0.8	707.6	2.86
71	Exit Long	Short	22/05/2023	24719.21	1	-370.22	-1.48	-19446.59	-0.04	153.27	0.61	417.62	1.66
71	Entry Long	Long	21/05/2023	25089.43	1	-370.22	-1.48	-19446.59	-0.04	153.27	0.61	417.62	1.66
70	Exit Short	Long	21/05/2023	25089.43	1	743.99	2.88	-19076.37	0.08	2039.35	7.89	180.13	0.7
70	Entry Short	Short	08/05/2023	25833.42	1	743.99	2.88	-19076.37	0.08	2039.35	7.89	180.13	0.7
69	Exit Long	Short	08/05/2023	25833.42	1	-212.86	-0.82	-19820.36	-0.02	1022.97	3.93	605.29	2.32
69	Entry Long	Long	03/05/2023	26046.28	1	-212.86	-0.82	-19820.36	-0.02	1022.97	3.93	605.29	2.32
68	Exit Short	Long	03/05/2023	26046.28	1	-458.49	-1.79	-19607.5	-0.05	199.96	0.78	683.29	2.67
68	Entry Short	Short	02/05/2023	25587.79	1	-458.49	-1.79	-19607.5	-0.05	199.96	0.78	683.29	2.67
67	Exit Long	Short	02/05/2023	25587.79	1	-1149.61	-4.3	-19149.01	-0.12	438.92	1.64	1523.01	5.7
67	Entry Long	Long	28/04/2023	26737.4	1	-1149.61	-4.3	-19149.01	-0.12	438.92	1.64	1523.01	5.7
66	Exit Short	Long	28/04/2023	26737.4	1	-414.16	-1.57	-17999.4	-0.04	1891.21	7.18	848.07	3.22
66	Entry Short	Short	20/04/2023	26323.24	1	-414.16	-1.57	-17999.4	-0.04	1891.21	7.18	848.07	3.22
65	Exit Long	Short	20/04/2023	26323.24	1	339.93	1.31	-17585.24	0.03	2066.78	7.95	83.31	0.32
65	Entry Long	Long	10/04/2023	25983.31	1	339.93	1.31	-17585.24	0.03	2066.78	7.95	83.31	0.32
64	Exit Short	Long	10/04/2023	25983.31	1	-285.99	-1.11	-17925.17	-0.03	222.56	0.87	502.66	1.96
64	Entry Short	Short	07/04/2023	25697.32	1	-285.99	-1.11	-17925.17	-0.03	222.56	0.87	502.66	1.96
63	Exit Long	Short	07/04/2023	25697.32	1	-143.26	-0.55	-17639.18	-0.01	14.42	0.06	410.83	1.59
63	Entry Long	Long	06/04/2023	25840.58	1	-143.26	-0.55	-17639.18	-0.01	14.42	0.06	410.83	1.59
62	Exit Short	Long	06/04/2023	25840.58	1	-340.04	-1.33	-17495.92	-0.03	116.62	0.46	749.46	2.94
62	Entry Short	Short	04/04/2023	25500.54	1	-340.04	-1.33	-17495.92	-0.03	116.62	0.46	749.46	2.94
61	Exit Long	Short	04/04/2023	25500.54	1	-653.63	-2.5	-17155.88	-0.07	767.42	2.93	1164.17	4.45
61	Entry Long	Long	30/03/2023	26154.17	1	-653.63	-2.5	-17155.88	-0.07	767.42	2.93	1164.17	4.45
60	Exit Short	Long	30/03/2023	26154.17	1	-1028.69	-4.09	-16502.25	-0.1	492.55	1.96	1307.22	5.2
60	Entry Short	Short	28/03/2023	25125.48	1	-1028.69	-4.09	-16502.25	-0.1	492.55	1.96	1307.22	5.2
59	Exit Long	Short	28/03/2023	25125.48	1	2859.78	12.84	-15473.56	0.29	4557.13	20.47	70.11	0.31
59	Entry Long	Long	14/03/2023	22265.7	1	2859.78	12.84	-15473.56	0.29	4557.13	20.47	70.11	0.31
58	Exit Short	Long	14/03/2023	22265.7	1	-117.33	-0.53	-18333.34	-0.01	3685.37	16.64	587.63	2.65
58	Entry Short	Short	28/02/2023	22148.37	1	-117.33	-0.53	-18333.34	-0.01	3685.37	16.64	587.63	2.65
57	Exit Long	Short	28/02/2023	22148.37	1	-178.07	-0.8	-18216.01	-0.02	222.54	1	522.81	2.34
57	Entry Long	Long	27/02/2023	22326.44	1	-178.07	-0.8	-18216.01	-0.02	222.54	1	522.81	2.34
56	Exit Short	Long	27/02/2023	22326.44	1	-342.51	-1.56	-18037.94	-0.03	383.93	1.75	478.99	2.18
56	Entry Short	Short	25/02/2023	21983.93	1	-342.51	-1.56	-18037.94	-0.03	383.93	1.75	478.99	2.18
55	Exit Long	Short	25/02/2023	21983.93	1	-753.12	-3.31	-17695.43	-0.08	922.25	4.06	1079.05	4.75
55	Entry Long	Long	16/02/2023	22737.05	1	-753.12	-3.31	-17695.43	-0.08	922.25	4.06	1079.05	4.75
54	Exit Short	Long	16/02/2023	22737.05	1	-2440.11	-12.02	-16942.31	-0.25	349.47	1.72	2501.91	12.33
54	Entry Short	Short	10/02/2023	20296.94	1	-2440.11	-12.02	-16942.31	-0.25	349.47	1.72	2501.91	12.33

53	Exit Long	Short	10/02/2023	20296.94	1	-1356.97	-6.27	-14502.2	-0.14	171.72	0.79	1442.05	6.66
53	Entry Long	Long	08/02/2023	21653.91	1	-1356.97	-6.27	-14502.2	-0.14	171.72	0.79	1442.05	6.66
52	Exit Short	Long	08/02/2023	21653.91	1	-390.03	-1.83	-13145.23	-0.04	292.61	1.38	450.41	2.12
52	Entry Short	Short	06/02/2023	21263.88	1	-390.03	-1.83	-13145.23	-0.04	292.61	1.38	450.41	2.12
51	Exit Long	Short	06/02/2023	21263.88	1	5467.63	34.61	-12755.2	0.56	6315.75	39.98	24.77	0.16
51	Entry Long	Long	04/01/2023	15796.25	1	5467.63	34.61	-12755.2	0.56	6315.75	39.98	24.77	0.16
50	Exit Short	Long	04/01/2023	15796.25	1	-86.97	-0.55	-18222.83	-0.01	406.02	2.58	297.72	1.9
50	Entry Short	Short	17/12/2022	15709.28	1	-86.97	-0.55	-18222.83	-0.01	406.02	2.58	297.72	1.9
49	Exit Long	Short	17/12/2022	15709.28	1	-603.43	-3.7	-18135.86	-0.06	887.29	5.44	699.91	4.29
49	Entry Long	Long	09/12/2022	16312.71	1	-603.43	-3.7	-18135.86	-0.06	887.29	5.44	699.91	4.29
48	Exit Short	Long	09/12/2022	16312.71	1	-290.21	-1.81	-17532.43	-0.03	102.5	0.64	346.63	2.16
48	Entry Short	Short	08/12/2022	16022.5	1	-290.21	-1.81	-17532.43	-0.03	102.5	0.64	346.63	2.16
47	Exit Long	Short	08/12/2022	16022.5	1	102.83	0.65	-17242.22	0.01	647.44	4.07	20.41	0.13
47	Entry Long	Long	30/11/2022	15919.67	1	102.83	0.65	-17242.22	0.01	647.44	4.07	20.41	0.13
46	Exit Short	Long	30/11/2022	15919.67	1	4640.2	22.57	-17345.05	0.47	5449.87	26.51	78.88	0.38
46	Entry Short	Short	08/11/2022	20559.87	1	4640.2	22.57	-17345.05	0.47	5449.87	26.51	78.88	0.38
45	Exit Long	Short	08/11/2022	20559.87	1	378.05	1.87	-21985.25	0.04	1380.53	6.84	57.01	0.28
45	Entry Long	Long	26/10/2022	20181.82	1	378.05	1.87	-21985.25	0.04	1380.53	6.84	57.01	0.28
44	Exit Short	Long	26/10/2022	20181.82	1	-626.25	-3.2	-22363.3	-0.06	89.18	0.46	923.43	4.72
44	Entry Short	Short	25/10/2022	19555.57	1	-626.25	-3.2	-22363.3	-0.06	89.18	0.46	923.43	4.72
43	Exit Long	Short	25/10/2022	19555.57	1	-293.52	-1.48	-21737.05	-0.03	32.33	0.16	413.14	2.08
43	Entry Long	Long	24/10/2022	19849.09	1	-293.52	-1.48	-21737.05	-0.03	32.33	0.16	413.14	2.08
42	Exit Short	Long	24/10/2022	19849.09	1	83.75	0.42	-21443.53	0.01	1093.81	5.49	417.4	2.09
42	Entry Short	Short	09/10/2022	19932.84	1	83.75	0.42	-21443.53	0.01	1093.81	5.49	417.4	2.09
41	Exit Long	Short	09/10/2022	19932.84	1	-37.56	-0.19	-21527.28	0	674.81	3.38	200.5	1
41	Entry Long	Long	04/10/2022	19970.4	1	-37.56	-0.19	-21527.28	0	674.81	3.38	200.5	1
40	Exit Short	Long	04/10/2022	19970.4	1	-513.85	-2.64	-21489.72	-0.05	81.43	0.42	600.14	3.08
40	Entry Short	Short	03/10/2022	19456.55	1	-513.85	-2.64	-21489.72	-0.05	81.43	0.42	600.14	3.08
39	Exit Long	Short	03/10/2022	19456.55	1	-527.96	-2.64	-20975.87	-0.05	1149.16	5.75	666.69	3.34
39	Entry Long	Long	27/09/2022	19984.51	1	-527.96	-2.64	-20975.87	-0.05	1149.16	5.75	666.69	3.34
38	Exit Short	Long	27/09/2022	19984.51	1	215.24	1.07	-20447.91	0.02	1899.75	9.4	354.55	1.76
38	Entry Short	Short	14/09/2022	20199.75	1	215.24	1.07	-20447.91	0.02	1899.75	9.4	354.55	1.76
37	Exit Long	Short	14/09/2022	20199.75	1	-1068.29	-5.02	-20663.15	-0.11	1183.73	5.57	1328.04	6.24
37	Entry Long	Long	10/09/2022	21268.04	1	-1068.29	-5.02	-20663.15	-0.11	1183.73	5.57	1328.04	6.24
36	Exit Short	Long	10/09/2022	21268.04	1	1670.33	7.28	-19594.86	0.17	4183.37	18.24	297.88	1.3
36	Entry Short	Short	18/08/2022	22938.37	1	1670.33	7.28	-19594.86	0.17	4183.37	18.24	297.88	1.3
35	Exit Long	Short	18/08/2022	22938.37	1	-328.15	-1.41	-21265.19	-0.03	1335.9	5.74	482.52	2.07
35	Entry Long	Long	11/08/2022	23266.52	1	-328.15	-1.41	-21265.19	-0.03	1335.9	5.74	482.52	2.07
34	Exit Short	Long	11/08/2022	23266.52	1	-584.41	-2.58	-20937.04	-0.06	464.11	2.05	842.89	3.72
34	Entry Short	Short	10/08/2022	22682.11	1	-584.41	-2.58	-20937.04	-0.06	464.11	2.05	842.89	3.72
33	Exit Long	Short	10/08/2022	22682.11	1	-205.89	-0.9	-20352.63	-0.02	884.39	3.86	533.64	2.33
33	Entry Long	Long	06/08/2022	22888	1	-205.89	-0.9	-20352.63	-0.02	884.39	3.86	533.64	2.33
32	Exit Short	Long	06/08/2022	22888	1	-807.34	-3.66	-20146.74	-0.08	20.16	0.09	865.5	3.92
32	Entry Short	Short	05/08/2022	22080.66	1	-807.34	-3.66	-20146.74	-0.08	20.16	0.09	865.5	3.92
31	Exit Long	Short	05/08/2022	22080.66	1	-426.14	-1.89	-19339.4	-0.04	1626.18	7.23	626.8	2.78
31	Entry Long	Long	28/07/2022	22506.8	1	-426.14	-1.89	-19339.4	-0.04	1626.18	7.23	626.8	2.78
30	Exit Short	Long	28/07/2022	22506.8	1	-1664.13	-7.98	-18913.26	-0.17	461.57	2.21	1794.77	8.61
30	Entry Short	Short	26/07/2022	20842.67	1	-1664.13	-7.98	-18913.26	-0.17	461.57	2.21	1794.77	8.61
29	Exit Long	Short	26/07/2022	20842.67	1	688.03	3.41	-17249.13	0.07	3643.36	18.08	554.64	2.75
29	Entry Long	Long	14/07/2022	20154.64	1	688.03	3.41	-17249.13	0.07	3643.36	18.08	554.64	2.75
28	Exit Short	Long	14/07/2022	20154.64	1	-898.61	-4.67	-17937.16	-0.09	378.03	1.96	980.23	5.09
28	Entry Short	Short	13/07/2022	19256.03	1	-898.61	-4.67	-17937.16	-0.09	378.03	1.96	980.23	5.09
27	Exit Long	Short	13/07/2022	19256.03	1	-113.78	-0.59	-17038.55	-0.01	2708.87	13.99	635.65	3.28
27	Entry Long	Long	05/07/2022	19369.81	1	-113.78	-0.59	-17038.55	-0.01	2708.87	13.99	635.65	3.28
26	Exit Short	Long	05/07/2022	19369.81	1	525.6	2.64	-16924.77	0.05	2131.41	10.71	439.59	2.21
26	Entry Short	Short	27/06/2022	19895.41	1	525.6	2.64	-16924.77	0.05	2131.41	10.71	439.59	2.21
25	Exit Long	Short	27/06/2022	19895.41	1	-462.38	-2.27	-17450.37	-0.05	378.21	1.86	517.13	2.54

25	Entry Long	Long	26/06/2022	20357.79	1	-462.38	-2.27	-17450.37	-0.05	378.21	1.86	517.13	2.54
24	Exit Short	Long	26/06/2022	20357.79	1	7278.62	26.34	-16987.99	0.75	10860.11	39.3	335.36	1.21
24	Entry Short	Short	11/06/2022	27636.41	1	7278.62	26.34	-16987.99	0.75	10860.11	39.3	335.36	1.21
23	Exit Long	Short	11/06/2022	27636.41	1	-237.95	-0.85	-24266.61	-0.02	1785.43	6.41	542.03	1.94
23	Entry Long	Long	06/06/2022	27874.36	1	-237.95	-0.85	-24266.61	-0.02	1785.43	6.41	542.03	1.94
22	Exit Short	Long	06/06/2022	27874.36	1	-159.07	-0.57	-24028.66	-0.02	211.41	0.76	429.64	1.55
22	Entry Short	Short	04/06/2022	27715.29	1	-159.07	-0.57	-24028.66	-0.02	211.41	0.76	429.64	1.55
21	Exit Long	Short	04/06/2022	27715.29	1	-1722.8	-5.85	-23869.59	-0.18	732.97	2.49	2153.09	7.31
21	Entry Long	Long	31/05/2022	29438.09	1	-1722.8	-5.85	-23869.59	-0.18	732.97	2.49	2153.09	7.31
20	Exit Short	Long	31/05/2022	29438.09	1	-2222.45	-8.17	-22146.79	-0.23	1049.63	3.86	2620.24	9.63
20	Entry Short	Short	24/05/2022	27215.64	1	-2222.45	-8.17	-22146.79	-0.23	1049.63	3.86	2620.24	9.63
19	Exit Long	Short	24/05/2022	27215.64	1	-1396.47	-4.88	-19924.34	-0.14	261.78	0.91	1601.11	5.6
19	Entry Long	Long	23/05/2022	28612.11	1	-1396.47	-4.88	-19924.34	-0.14	261.78	0.91	1601.11	5.6
18	Exit Short	Long	23/05/2022	28612.11	1	6056.39	17.47	-18527.87	0.62	10442.2	30.12	115.36	0.33
18	Entry Short	Short	06/05/2022	34668.5	1	6056.39	17.47	-18527.87	0.62	10442.2	30.12	115.36	0.33
17	Exit Long	Short	06/05/2022	34668.5	1	-2692.57	-7.21	-24584.26	-0.28	163.57	0.44	3544.65	9.49
17	Entry Long	Long	05/05/2022	37361.07	1	-2692.57	-7.21	-24584.26	-0.28	163.57	0.44	3544.65	9.49
16	Exit Short	Long	05/05/2022	37361.07	1	-740.51	-2.02	-21891.69	-0.08	1118.56	3.05	1111.74	3.04
16	Entry Short	Short	30/04/2022	36620.56	1	-740.51	-2.02	-21891.69	-0.08	1118.56	3.05	1111.74	3.04
15	Exit Long	Short	30/04/2022	36620.56	1	-1216.36	-3.21	-21151.18	-0.12	80.98	0.21	1646.92	4.35
15	Entry Long	Long	29/04/2022	37836.92	1	-1216.36	-3.21	-21151.18	-0.12	80.98	0.21	1646.92	4.35
14	Exit Short	Long	29/04/2022	37836.92	1	-2009.38	-5.61	-19934.82	-0.2	215.34	0.6	2635.61	7.36
14	Entry Short	Short	27/04/2022	35827.54	1	-2009.38	-5.61	-19934.82	-0.2	215.34	0.6	2635.61	7.36
13	Exit Long	Short	27/04/2022	35827.54	1	-1922.58	-5.09	-17925.44	-0.2	438.07	1.16	2312.12	6.12
13	Entry Long	Long	26/04/2022	37750.12	1	-1922.58	-5.09	-17925.44	-0.2	438.07	1.16	2312.12	6.12
12	Exit Short	Long	26/04/2022	37750.12	1	-380.15	-1.02	-16002.86	-0.04	1803.97	4.83	549.48	1.47
12	Entry Short	Short	22/04/2022	37369.97	1	-380.15	-1.02	-16002.86	-0.04	1803.97	4.83	549.48	1.47
11	Exit Long	Short	22/04/2022	37369.97	1	-1069.85	-2.78	-15622.71	-0.11	1038.78	2.7	1714.41	4.46
11	Entry Long	Long	20/04/2022	38439.82	1	-1069.85	-2.78	-15622.71	-0.11	1038.78	2.7	1714.41	4.46
10	Exit Short	Long	20/04/2022	38439.82	1	1173.21	2.96	-14552.86	0.12	3873.57	9.78	868.82	2.19
10	Entry Short	Short	07/04/2022	39613.03	1	1173.21	2.96	-14552.86	0.12	3873.57	9.78	868.82	2.19
9	Exit Long	Short	07/04/2022	39613.03	1	2306.83	6.18	-15726.07	0.23	6601.33	17.69	1026.17	2.75
9	Entry Long	Long	17/03/2022	37306.2	1	2306.83	6.18	-15726.07	0.23	6601.33	17.69	1026.17	2.75
8	Exit Short	Long	17/03/2022	37306.2	1	-1491.19	-4.16	-18032.9	-0.15	1415.01	3.95	2184.96	6.1
8	Entry Short	Short	11/03/2022	35815.01	1	-1491.19	-4.16	-18032.9	-0.15	1415.01	3.95	2184.96	6.1
7	Exit Long	Short	11/03/2022	35815.01	1	-2082.05	-5.49	-16541.71	-0.21	142.69	0.38	3017.06	7.96
7	Entry Long	Long	10/03/2022	37897.06	1	-2082.05	-5.49	-16541.71	-0.21	142.69	0.38	3017.06	7.96
6	Exit Short	Long	10/03/2022	37897.06	1	-2560.57	-7.25	-14459.66	-0.26	1071.73	3.03	3463.51	9.8
6	Entry Short	Short	07/03/2022	35336.49	1	-2560.57	-7.25	-14459.66	-0.26	1071.73	3.03	3463.51	9.8
5	Exit Long	Short	07/03/2022	35336.49	1	-3165.63	-8.22	-11899.09	-0.32	2395.02	6.22	3612.39	9.38
5	Entry Long	Long	01/03/2022	38502.12	1	-3165.63	-8.22	-11899.09	-0.32	2395.02	6.22	3612.39	9.38
4	Exit Short	Long	01/03/2022	38502.12	1	-2828.17	-7.93	-8733.46	-0.28	5129.74	14.38	3716.5	10.42
4	Entry Short	Short	18/02/2022	35673.95	1	-2828.17	-7.93	-8733.46	-0.28	5129.74	14.38	3716.5	10.42
3	Exit Long	Short	18/02/2022	35673.95	1	-647.88	-1.78	-5905.29	-0.07	3623.17	9.98	1071.83	2.95
3	Entry Long	Long	05/02/2022	36321.83	1	-647.88	-1.78	-5905.29	-0.07	3623.17	9.98	1071.83	2.95
2	Exit Short	Long	05/02/2022	36321.83	1	-3637.71	-11.13	-5257.41	-0.36	939.11	2.87	3777.09	11.56
2	Entry Short	Short	03/02/2022	32684.12	1	-3637.71	-11.13	-5257.41	-0.36	939.11	2.87	3777.09	11.56
1	Exit Long	Short	03/02/2022	32684.12	1	-1619.7	-4.72	-1619.7	-0.16	666.15	1.94	1903.82	5.55
1	Entry Long	Long	30/01/2022	34303.82	1	-1619.7	-4.72	-1619.7	-0.16	666.15	1.94	1903.82	5.55

Appendix 17. List of trades (Exponential Moving Average)

LIST OF TRADES (EMA)

Trade #	Type	Signal	Date/Time	Price EUR	Contracts	Profit EUR	Profit %	Cum. Profit EUR	Cum. Profit %	Run-up EUR	Run-up %	Drawdown EUR	Drawdown %
125	Exit Long	Close entry(s) order Long	13/01/2024	39150	1	-941.14	-2.35	-28912.02	-0.1	4796.6	11.96	2741.39	6.84
125	Entry Long	Long	02/01/2024	40091	1	-941.14	-2.35	-28912.02	-0.1	4796.6	11.96	2741.39	6.84
124	Exit Short	Long	02/01/2024	40091	1	-1581.7	-4.11	-27970.88	-0.16	965.42	2.51	1590.27	4.13
124	Entry Short	Short	29/12/2023	38510	1	-1581.7	-4.11	-27970.88	-0.16	965.42	2.51	1590.27	4.13
123	Exit Long	Short	29/12/2023	38510	1	-646.82	-1.65	-26389.22	-0.07	293.21	0.75	998.14	2.55
123	Entry Long	Long	28/12/2023	39157	1	-646.82	-1.65	-26389.22	-0.07	293.21	0.75	998.14	2.55
122	Exit Short	Long	28/12/2023	39157	1	-623.14	-1.62	-25742.4	-0.06	333.41	0.87	852.59	2.21
122	Entry Short	Short	27/12/2023	38533	1	-623.14	-1.62	-25742.4	-0.06	333.41	0.87	852.59	2.21
121	Exit Long	Short	27/12/2023	38533	1	-1049.7	-2.65	-25119.26	-0.11	16.65	0.04	1827.71	4.62
121	Entry Long	Long	26/12/2023	39583	1	-1049.7	-2.65	-25119.26	-0.11	16.65	0.04	1827.71	4.62
120	Exit Short	Long	26/12/2023	39583	1	-425.51	-1.09	-24069.57	-0.04	204.7	0.52	710.29	1.81
120	Entry Short	Short	25/12/2023	39158	1	-425.51	-1.09	-24069.57	-0.04	204.7	0.52	710.29	1.81
119	Exit Long	Short	25/12/2023	39158	1	-733.65	-1.84	-23644.06	-0.08	498.76	1.25	1076.29	2.7
119	Entry Long	Long	21/12/2023	39891	1	-733.65	-1.84	-23644.06	-0.08	498.76	1.25	1076.29	2.7
118	Exit Short	Long	21/12/2023	39891	1	-1376	-3.57	-22910.41	-0.14	26.8	0.07	1912.85	4.97
118	Entry Short	Short	20/12/2023	38515	1	-1376	-3.57	-22910.41	-0.14	26.8	0.07	1912.85	4.97
117	Exit Long	Short	20/12/2023	38515	1	-543.23	-1.39	-21534.42	-0.06	755.01	1.93	982.12	2.51
117	Entry Long	Long	19/12/2023	39058	1	-543.23	-1.39	-21534.42	-0.06	755.01	1.93	982.12	2.51
116	Exit Short	Long	19/12/2023	39058	1	-1082.4	-2.85	-20991.19	-0.11	834.83	2.2	1163.52	3.06
116	Entry Short	Short	18/12/2023	37976	1	-1082.4	-2.85	-20991.19	-0.11	834.83	2.2	1163.52	3.06
115	Exit Long	Short	18/12/2023	37976	1	-851.52	-2.19	-19908.79	-0.09	120.47	0.31	957.27	2.47
115	Entry Long	Long	17/12/2023	38828	1	-851.52	-2.19	-19908.79	-0.09	120.47	0.31	957.27	2.47
114	Exit Short	Long	17/12/2023	38828	1	-304.8	-0.79	-19057.27	-0.03	285.73	0.74	700.13	1.82
114	Entry Short	Short	16/12/2023	38523	1	-304.8	-0.79	-19057.27	-0.03	285.73	0.74	700.13	1.82
113	Exit Long	Short	16/12/2023	38523	1	-904.77	-2.29	-18752.47	-0.09	251.4	0.64	1427.57	3.62
113	Entry Long	Long	14/12/2023	39428	1	-904.77	-2.29	-18752.47	-0.09	251.4	0.64	1427.57	3.62
112	Exit Short	Long	14/12/2023	39428	1	-1114.7	-2.91	-17847.7	-0.11	700.07	1.83	1676.14	4.37
112	Entry Short	Short	12/12/2023	38313	1	-1114.7	-2.91	-17847.7	-0.11	700.07	1.83	1676.14	4.37
111	Exit Long	Short	12/12/2023	38313	1	3937.48	11.45	-16732.99	0.4	7207.27	20.97	772.38	2.25
111	Entry Long	Long	23/11/2023	34375	1	3937.48	11.45	-16732.99	0.4	7207.27	20.97	772.38	2.25
110	Exit Short	Long	23/11/2023	34375	1	-1603.9	-4.89	-20670.47	-0.16	94.02	0.29	2012.77	6.14
110	Entry Short	Short	22/11/2023	32772	1	-1603.9	-4.89	-20670.47	-0.16	94.02	0.29	2012.77	6.14
109	Exit Long	Short	22/11/2023	32772	1	-837.34	-2.49	-19066.6	-0.09	934.15	2.78	847.51	2.52
109	Entry Long	Long	18/11/2023	33609	1	-837.34	-2.49	-19066.6	-0.09	934.15	2.78	847.51	2.52
108	Exit Short	Long	18/11/2023	33609	1	-271.03	-0.81	-18229.26	-0.03	307.21	0.92	537.94	1.61
108	Entry Short	Short	17/11/2023	33338	1	-271.03	-0.81	-18229.26	-0.03	307.21	0.92	537.94	1.61
107	Exit Long	Short	17/11/2023	33338	1	-1619	-4.63	-17958.23	-0.16	33.27	0.1	2246.81	6.43
107	Entry Long	Long	16/11/2023	34957	1	-1619	-4.63	-17958.23	-0.16	33.27	0.1	2246.81	6.43
106	Exit Short	Long	16/11/2023	34957	1	-2250.5	-6.88	-16339.24	-0.23	172.43	0.53	2315.24	7.08
106	Entry Short	Short	15/11/2023	32706	1	-2250.5	-6.88	-16339.24	-0.23	172.43	0.53	2315.24	7.08
105	Exit Long	Short	15/11/2023	32706	1	6876.41	26.62	-14088.72	0.7	9644.4	37.34	50.24	0.19
105	Entry Long	Long	16/10/2023	25830	1	6876.41	26.62	-14088.72	0.7	9644.4	37.34	50.24	0.19
104	Exit Short	Long	16/10/2023	25830	1	-6.31	-0.02	-20965.13	0	754.64	2.92	114.26	0.44
104	Entry Short	Short	11/10/2023	25824	1	-6.31	-0.02	-20965.13	0	754.64	2.92	114.26	0.44
103	Exit Long	Short	11/10/2023	25824	1	745.71	2.97	-20958.82	0.08	2118.41	8.45	41.4	0.17
103	Entry Long	Long	28/09/2023	25078	1	745.71	2.97	-20958.82	0.08	2118.41	8.45	41.4	0.17
102	Exit Short	Long	28/09/2023	25078	1	-423.53	-1.72	-21704.53	-0.04	239.33	0.97	793.36	3.22
102	Entry Short	Short	25/09/2023	24654	1	-423.53	-1.72	-21704.53	-0.04	239.33	0.97	793.36	3.22
101	Exit Long	Short	25/09/2023	24654	1	213.36	0.87	-21281	0.02	1291.29	5.28	95.42	0.39
101	Entry Long	Long	14/09/2023	24441	1	213.36	0.87	-21281	0.02	1291.29	5.28	95.42	0.39
100	Exit Short	Long	14/09/2023	24441	1	-221.86	-0.92	-21494.36	-0.02	1044.11	4.31	512.01	2.11
100	Entry Short	Short	09/09/2023	24219	1	-221.86	-0.92	-21494.36	-0.02	1044.11	4.31	512.01	2.11
99	Exit Long	Short	09/09/2023	24219	1	-334.94	-1.36	-21272.5	-0.03	137.45	0.56	578.68	2.36
99	Entry Long	Long	08/09/2023	24554	1	-334.94	-1.36	-21272.5	-0.03	137.45	0.56	578.68	2.36
98	Exit Short	Long	08/09/2023	24554	1	-644.33	-2.69	-20937.56	-0.07	440.72	1.84	803.61	3.36

98	Entry Short	Short	01/09/2023	23910	1	-644.33	-2.69	-20937.56	-0.07	440.72	1.84	803.61	3.36
97	Exit Long	Short	01/09/2023	23910	1	-1574.5	-6.18	-20293.23	-0.16	45.37	0.18	1811.86	7.11
97	Entry Long	Long	30/08/2023	25484	1	-1574.5	-6.18	-20293.23	-0.16	45.37	0.18	1811.86	7.11
96	Exit Short	Long	30/08/2023	25484	1	1257.06	4.7	-18718.76	0.13	3628.25	13.57	60.79	0.23
96	Entry Short	Short	16/08/2023	26741	1	1257.06	4.7	-18718.76	0.13	3628.25	13.57	60.79	0.23
95	Exit Long	Short	16/08/2023	26741	1	-406.34	-1.5	-19975.82	-0.04	270.18	1	555.14	2.04
95	Entry Long	Long	09/08/2023	27148	1	-406.34	-1.5	-19975.82	-0.04	270.18	1	555.14	2.04
94	Exit Short	Long	09/08/2023	27148	1	-496.44	-1.86	-19569.48	-0.05	571.15	2.14	918.85	3.45
94	Entry Short	Short	03/08/2023	26651	1	-496.44	-1.86	-19569.48	-0.05	571.15	2.14	918.85	3.45
93	Exit Long	Short	03/08/2023	26651	1	-324.1	-1.2	-19073.04	-0.03	319.12	1.18	565.7	2.1
93	Entry Long	Long	02/08/2023	26975	1	-324.1	-1.2	-19073.04	-0.03	319.12	1.18	565.7	2.1
92	Exit Short	Long	02/08/2023	26975	1	-587.66	-2.23	-18748.94	-0.06	329.58	1.25	600	2.27
92	Entry Short	Short	25/07/2023	26388	1	-587.66	-2.23	-18748.94	-0.06	329.58	1.25	600	2.27
91	Exit Long	Short	25/07/2023	26388	1	-665.27	-2.46	-18161.28	-0.07	9.02	0.03	983.86	3.64
91	Entry Long	Long	24/07/2023	27053	1	-665.27	-2.46	-18161.28	-0.07	9.02	0.03	983.86	3.64
90	Exit Short	Long	24/07/2023	27053	1	-6.34	-0.02	-17496.01	0	700.64	2.59	245.57	0.91
90	Entry Short	Short	15/07/2023	27047	1	-6.34	-0.02	-17496.01	0	700.64	2.59	245.57	0.91
89	Exit Long	Short	15/07/2023	27047	1	-980.17	-3.5	-17489.67	-0.1	127.15	0.45	1386.69	4.95
89	Entry Long	Long	14/07/2023	28027	1	-980.17	-3.5	-17489.67	-0.1	127.15	0.45	1386.69	4.95
88	Exit Short	Long	14/07/2023	28027	1	-753.82	-2.76	-16509.5	-0.08	127.7	0.47	1076.07	3.95
88	Entry Short	Short	13/07/2023	27273	1	-753.82	-2.76	-16509.5	-0.08	127.7	0.47	1076.07	3.95
87	Exit Long	Short	13/07/2023	27273	1	-355.73	-1.29	-15755.68	-0.04	428.85	1.55	496.76	1.8
87	Entry Long	Long	11/07/2023	27629	1	-355.73	-1.29	-15755.68	-0.04	428.85	1.55	496.76	1.8
86	Exit Short	Long	11/07/2023	27629	1	-117.79	-0.43	-15399.95	-0.01	172.29	0.63	711.65	2.59
86	Entry Short	Short	10/07/2023	27511	1	-117.79	-0.43	-15399.95	-0.01	172.29	0.63	711.65	2.59
85	Exit Long	Short	10/07/2023	27511	1	-174.33	-0.63	-15282.16	-0.02	89.86	0.32	273.65	0.99
85	Entry Long	Long	08/07/2023	27685	1	-174.33	-0.63	-15282.16	-0.02	89.86	0.32	273.65	0.99
84	Exit Short	Long	08/07/2023	27685	1	-220.79	-0.8	-15107.83	-0.02	169.35	0.62	445.02	1.62
84	Entry Short	Short	07/07/2023	27464	1	-220.79	-0.8	-15107.83	-0.02	169.35	0.62	445.02	1.62
83	Exit Long	Short	07/07/2023	27464	1	2891.6	11.77	-14887.04	0.29	4425.48	18.01	156.25	0.64
83	Entry Long	Long	20/06/2023	24573	1	2891.6	11.77	-14887.04	0.29	4425.48	18.01	156.25	0.64
82	Exit Short	Long	20/06/2023	24573	1	18.5	0.08	-17778.64	0	1809	7.36	286.37	1.16
82	Entry Short	Short	08/06/2023	24591	1	18.5	0.08	-17778.64	0	1809	7.36	286.37	1.16
81	Exit Long	Short	08/06/2023	24591	1	-862.36	-3.39	-17797.14	-0.09	135.54	0.53	1033.61	4.06
81	Entry Long	Long	07/06/2023	25454	1	-862.36	-3.39	-17797.14	-0.09	135.54	0.53	1033.61	4.06
80	Exit Short	Long	07/06/2023	25454	1	-1422	-5.92	-16934.78	-0.14	278.43	1.16	1538.32	6.4
80	Entry Short	Short	06/06/2023	24032	1	-1422	-5.92	-16934.78	-0.14	278.43	1.16	1538.32	6.4
79	Exit Long	Short	06/06/2023	24032	1	-1312.8	-5.18	-15512.82	-0.13	34.83	0.14	1621.23	6.4
79	Entry Long	Long	05/06/2023	25344	1	-1312.8	-5.18	-15512.82	-0.13	34.83	0.14	1621.23	6.4
78	Exit Short	Long	05/06/2023	25344	1	-57.69	-0.23	-14200.03	-0.01	105.17	0.42	359.17	1.42
78	Entry Short	Short	04/06/2023	25287	1	-57.69	-0.23	-14200.03	-0.01	105.17	0.42	359.17	1.42
77	Exit Long	Short	04/06/2023	25287	1	-165.25	-0.65	-14142.34	-0.02	80.79	0.32	303.51	1.19
77	Entry Long	Long	03/06/2023	25452	1	-165.25	-0.65	-14142.34	-0.02	80.79	0.32	303.51	1.19
76	Exit Short	Long	03/06/2023	25452	1	-516.75	-2.07	-13977.09	-0.05	287.33	1.15	566.75	2.27
76	Entry Short	Short	02/06/2023	24935	1	-516.75	-2.07	-13977.09	-0.05	287.33	1.15	566.75	2.27
75	Exit Long	Short	02/06/2023	24935	1	-113.85	-0.45	-13460.34	-0.01	1492.92	5.96	265.45	1.06
75	Entry Long	Long	28/05/2023	25049	1	-113.85	-0.45	-13460.34	-0.01	1492.92	5.96	265.45	1.06
74	Exit Short	Long	28/05/2023	25049	1	-574.32	-2.35	-13346.49	-0.06	389.89	1.59	670.95	2.74
74	Entry Short	Short	25/05/2023	24475	1	-574.32	-2.35	-13346.49	-0.06	389.89	1.59	670.95	2.74
73	Exit Long	Short	25/05/2023	24475	1	-797.69	-3.16	-12772.17	-0.08	7.04	0.03	1048.37	4.15
73	Entry Long	Long	24/05/2023	25272	1	-797.69	-3.16	-12772.17	-0.08	7.04	0.03	1048.37	4.15
72	Exit Short	Long	24/05/2023	25272	1	-390.42	-1.57	-11974.48	-0.04	360.55	1.45	544.76	2.19
72	Entry Short	Short	19/05/2023	24882	1	-390.42	-1.57	-11974.48	-0.04	360.55	1.45	544.76	2.19
71	Exit Long	Short	19/05/2023	24882	1	-398.81	-1.58	-11584.06	-0.04	128.75	0.51	794.24	3.14
71	Entry Long	Long	18/05/2023	25281	1	-398.81	-1.58	-11584.06	-0.04	128.75	0.51	794.24	3.14
70	Exit Short	Long	18/05/2023	25281	1	552.56	2.14	-11185.25	0.06	2039.35	7.89	180.13	0.7
70	Entry Short	Short	08/05/2023	25833	1	552.56	2.14	-11185.25	0.06	2039.35	7.89	180.13	0.7

69	Exit Long	Short	08/05/2023	25833	1	-402.59	-1.53	-11737.81	-0.04	833.24	3.18	450.56	1.72
69	Entry Long	Long	04/05/2023	26236	1	-402.59	-1.53	-11737.81	-0.04	833.24	3.18	450.56	1.72
68	Exit Short	Long	04/05/2023	26236	1	-648.22	-2.53	-11335.22	-0.07	199.96	0.78	854.58	3.34
68	Entry Short	Short	02/05/2023	25588	1	-648.22	-2.53	-11335.22	-0.07	199.96	0.78	854.58	3.34
67	Exit Long	Short	02/05/2023	25588	1	-1149.6	-4.3	-10687	-0.12	438.92	1.64	1523.01	5.7
67	Entry Long	Long	28/04/2023	26737	1	-1149.6	-4.3	-10687	-0.12	438.92	1.64	1523.01	5.7
66	Exit Short	Long	28/04/2023	26737	1	-414.16	-1.57	-9537.39	-0.04	1891.21	7.18	848.07	3.22
66	Entry Short	Short	20/04/2023	26323	1	-414.16	-1.57	-9537.39	-0.04	1891.21	7.18	848.07	3.22
65	Exit Long	Short	20/04/2023	26323	1	603.88	2.35	-9123.23	0.06	2330.73	9.06	289.61	1.13
65	Entry Long	Long	05/04/2023	25719	1	603.88	2.35	-9123.23	0.06	2330.73	9.06	289.61	1.13
64	Exit Short	Long	05/04/2023	25719	1	-218.82	-0.86	-9727.11	-0.02	116.62	0.46	577.29	2.26
64	Entry Short	Short	04/04/2023	25501	1	-218.82	-0.86	-9727.11	-0.02	116.62	0.46	577.29	2.26
63	Exit Long	Short	04/04/2023	25501	1	3234.84	14.53	-9508.29	0.33	4655.89	20.91	70.11	0.31
63	Entry Long	Long	14/03/2023	22266	1	3234.84	14.53	-9508.29	0.33	4655.89	20.91	70.11	0.31
62	Exit Short	Long	14/03/2023	22266	1	-114.8	-0.52	-12743.13	-0.01	3687.9	16.65	585.1	2.64
62	Entry Short	Short	03/03/2023	22151	1	-114.8	-0.52	-12743.13	-0.01	3687.9	16.65	585.1	2.64
61	Exit Long	Short	03/03/2023	22151	1	-1.23	-0.01	-12628.33	0	147.87	0.67	253.83	1.15
61	Entry Long	Long	02/03/2023	22152	1	-1.23	-0.01	-12628.33	0	147.87	0.67	253.83	1.15
60	Exit Short	Long	02/03/2023	22152	1	-3.76	-0.02	-12627.1	0	369.67	1.67	392.64	1.77
60	Entry Short	Short	28/02/2023	22148	1	-3.76	-0.02	-12627.1	0	369.67	1.67	392.64	1.77
59	Exit Long	Short	28/02/2023	22148	1	-178.07	-0.8	-12623.34	-0.02	222.54	1	522.81	2.34
59	Entry Long	Long	27/02/2023	22326	1	-178.07	-0.8	-12623.34	-0.02	222.54	1	522.81	2.34
58	Exit Short	Long	27/02/2023	22326	1	-342.51	-1.56	-12445.27	-0.03	383.93	1.75	478.99	2.18
58	Entry Short	Short	25/02/2023	21984	1	-342.51	-1.56	-12445.27	-0.03	383.93	1.75	478.99	2.18
57	Exit Long	Short	25/02/2023	21984	1	-753.12	-3.31	-12102.76	-0.08	922.25	4.06	1079.05	4.75
57	Entry Long	Long	16/02/2023	22737	1	-753.12	-3.31	-12102.76	-0.08	922.25	4.06	1079.05	4.75
56	Exit Short	Long	16/02/2023	22737	1	-2440.1	-12	-11349.64	-0.25	349.47	1.72	2501.91	12.33
56	Entry Short	Short	10/02/2023	20297	1	-2440.1	-12	-11349.64	-0.25	349.47	1.72	2501.91	12.33
55	Exit Long	Short	10/02/2023	20297	1	4500.69	28.49	-8909.53	0.46	6315.75	39.98	24.77	0.16
55	Entry Long	Long	04/01/2023	15796	1	4500.69	28.49	-8909.53	0.46	6315.75	39.98	24.77	0.16
54	Exit Short	Long	04/01/2023	15796	1	501.37	3.08	-13410.22	0.05	994.36	6.1	170.07	1.04
54	Entry Short	Short	16/12/2022	16298	1	501.37	3.08	-13410.22	0.05	994.36	6.1	170.07	1.04
53	Exit Long	Short	16/12/2022	16298	1	-15.09	-0.09	-13911.59	0	887.29	5.44	298.33	1.83
53	Entry Long	Long	09/12/2022	16313	1	-15.09	-0.09	-13911.59	0	887.29	5.44	298.33	1.83
52	Exit Short	Long	09/12/2022	16313	1	-290.21	-1.81	-13896.5	-0.03	102.5	0.64	346.63	2.16
52	Entry Short	Short	08/12/2022	16023	1	-290.21	-1.81	-13896.5	-0.03	102.5	0.64	346.63	2.16
51	Exit Long	Short	08/12/2022	16023	1	-205.36	-1.27	-13606.29	-0.02	261.14	1.61	289.13	1.78
51	Entry Long	Long	05/12/2022	16228	1	-205.36	-1.27	-13606.29	-0.02	261.14	1.61	289.13	1.78
50	Exit Short	Long	05/12/2022	16228	1	-187.3	-1.17	-13400.93	-0.02	0	0	281.19	1.75
50	Entry Short	Short	04/12/2022	16041	1	-187.3	-1.17	-13400.93	-0.02	0	0	281.19	1.75
49	Exit Long	Short	04/12/2022	16041	1	-189.87	-1.17	-13213.63	-0.02	49.57	0.31	214.43	1.32
49	Entry Long	Long	03/12/2022	16230	1	-189.87	-1.17	-13213.63	-0.02	49.57	0.31	214.43	1.32
48	Exit Short	Long	03/12/2022	16230	1	-92.43	-0.57	-13023.76	-0.01	90.24	0.56	104.27	0.65
48	Entry Short	Short	02/12/2022	16138	1	-92.43	-0.57	-13023.76	-0.01	90.24	0.56	104.27	0.65
47	Exit Long	Short	02/12/2022	16138	1	-332.17	-2.02	-12931.33	-0.03	79.81	0.48	434.52	2.64
47	Entry Long	Long	01/12/2022	16470	1	-332.17	-2.02	-12931.33	-0.03	79.81	0.48	434.52	2.64
46	Exit Short	Long	01/12/2022	16470	1	4089.7	19.89	-12599.16	0.42	5449.87	26.51	78.88	0.38
46	Entry Short	Short	08/11/2022	20560	1	4089.7	19.89	-12599.16	0.42	5449.87	26.51	78.88	0.38
45	Exit Long	Short	08/11/2022	20560	1	378.05	1.87	-16688.86	0.04	1380.53	6.84	57.01	0.28
45	Entry Long	Long	26/10/2022	20182	1	378.05	1.87	-16688.86	0.04	1380.53	6.84	57.01	0.28
44	Exit Short	Long	26/10/2022	20182	1	-626.25	-3.2	-17066.91	-0.06	89.18	0.46	923.43	4.72
44	Entry Short	Short	25/10/2022	19556	1	-626.25	-3.2	-17066.91	-0.06	89.18	0.46	923.43	4.72
43	Exit Long	Short	25/10/2022	19556	1	-293.52	-1.48	-16440.66	-0.03	32.33	0.16	413.14	2.08
43	Entry Long	Long	24/10/2022	19849	1	-293.52	-1.48	-16440.66	-0.03	32.33	0.16	413.14	2.08
42	Exit Short	Long	24/10/2022	19849	1	-246.06	-1.26	-16147.14	-0.03	433.03	2.21	359.97	1.84
42	Entry Short	Short	19/10/2022	19603	1	-246.06	-1.26	-16147.14	-0.03	433.03	2.21	359.97	1.84
41	Exit Long	Short	19/10/2022	19603	1	-257.15	-1.29	-15901.08	-0.03	130.14	0.66	458.18	2.31

41	Entry Long	Long	18/10/2022	19860	1	-257.15	-1.29	-15901.08	-0.03	130.14	0.66	458.18	2.31
40	Exit Short	Long	18/10/2022	19860	1	72.66	0.36	-15643.93	0.01	1093.81	5.49	417.4	2.09
40	Entry Short	Short	09/10/2022	19933	1	72.66	0.36	-15643.93	0.01	1093.81	5.49	417.4	2.09
39	Exit Long	Short	09/10/2022	19933	1	-37.56	-0.19	-15716.59	0	674.81	3.38	200.5	1
39	Entry Long	Long	04/10/2022	19970	1	-37.56	-0.19	-15716.59	0	674.81	3.38	200.5	1
38	Exit Short	Long	04/10/2022	19970	1	-147.77	-0.75	-15679.03	-0.02	504.81	2.55	234.06	1.18
38	Entry Short	Short	01/10/2022	19823	1	-147.77	-0.75	-15679.03	-0.02	504.81	2.55	234.06	1.18
37	Exit Long	Short	01/10/2022	19823	1	-161.88	-0.81	-15531.26	-0.02	1149.16	5.75	634.51	3.18
37	Entry Long	Long	27/09/2022	19985	1	-161.88	-0.81	-15531.26	-0.02	1149.16	5.75	634.51	3.18
36	Exit Short	Long	27/09/2022	19985	1	-473.13	-2.42	-15369.38	-0.05	252.06	1.29	498.61	2.56
36	Entry Short	Short	25/09/2022	19511	1	-473.13	-2.42	-15369.38	-0.05	252.06	1.29	498.61	2.56
35	Exit Long	Short	25/09/2022	19511	1	-382.4	-1.92	-14896.25	-0.04	16.12	0.08	495.47	2.49
35	Entry Long	Long	24/09/2022	19894	1	-382.4	-1.92	-14896.25	-0.04	16.12	0.08	495.47	2.49
34	Exit Short	Long	24/09/2022	19894	1	305.97	1.51	-14513.85	0.03	1899.75	9.4	354.55	1.76
34	Entry Short	Short	14/09/2022	20200	1	305.97	1.51	-14513.85	0.03	1899.75	9.4	354.55	1.76
33	Exit Long	Short	14/09/2022	20200	1	-1068.3	-5.02	-14819.82	-0.11	1183.73	5.57	1328.04	6.24
33	Entry Long	Long	10/09/2022	21268	1	-1068.3	-5.02	-14819.82	-0.11	1183.73	5.57	1328.04	6.24
32	Exit Short	Long	10/09/2022	21268	1	1670.33	7.28	-13751.53	0.17	4183.37	18.24	297.88	1.3
32	Entry Short	Short	18/08/2022	22938	1	1670.33	7.28	-13751.53	0.17	4183.37	18.24	297.88	1.3
31	Exit Long	Short	18/08/2022	22938	1	50.37	0.22	-15421.86	0.01	1714.42	7.49	670	2.93
31	Entry Long	Long	06/08/2022	22888	1	50.37	0.22	-15421.86	0.01	1714.42	7.49	670	2.93
30	Exit Short	Long	06/08/2022	22888	1	-807.34	-3.66	-15472.23	-0.08	20.16	0.09	865.5	3.92
30	Entry Short	Short	05/08/2022	22081	1	-807.34	-3.66	-15472.23	-0.08	20.16	0.09	865.5	3.92
29	Exit Long	Short	05/08/2022	22081	1	-426.14	-1.89	-14664.89	-0.04	1626.18	7.23	626.8	2.78
29	Entry Long	Long	28/07/2022	22507	1	-426.14	-1.89	-14664.89	-0.04	1626.18	7.23	626.8	2.78
28	Exit Short	Long	28/07/2022	22507	1	-1664.1	-7.98	-14238.75	-0.17	461.57	2.21	1794.77	8.61
28	Entry Short	Short	26/07/2022	20843	1	-1664.1	-7.98	-14238.75	-0.17	461.57	2.21	1794.77	8.61
27	Exit Long	Short	26/07/2022	20843	1	688.03	3.41	-12574.62	0.07	3643.36	18.08	554.64	2.75
27	Entry Long	Long	14/07/2022	20155	1	688.03	3.41	-12574.62	0.07	3643.36	18.08	554.64	2.75
26	Exit Short	Long	14/07/2022	20155	1	-294.19	-1.48	-13262.65	-0.03	982.45	4.95	375.81	1.89
26	Entry Short	Short	12/07/2022	19860	1	-294.19	-1.48	-13262.65	-0.03	982.45	4.95	375.81	1.89
25	Exit Long	Short	12/07/2022	19860	1	226.98	1.16	-12968.46	0.02	2445.21	12.45	398.22	2.03
25	Entry Long	Long	06/07/2022	19633	1	226.98	1.16	-12968.46	0.02	2445.21	12.45	398.22	2.03
24	Exit Short	Long	06/07/2022	19633	1	8548.42	30.33	-13195.44	0.87	11405.59	40.47	439.84	1.56
24	Entry Short	Short	09/06/2022	28182	1	8548.42	30.33	-13195.44	0.87	11405.59	40.47	439.84	1.56
23	Exit Long	Short	09/06/2022	28182	1	-1123.4	-3.83	-21743.86	-0.11	159.35	0.54	1972.92	6.73
23	Entry Long	Long	07/06/2022	29305	1	-1123.4	-3.83	-21743.86	-0.11	159.35	0.54	1972.92	6.73
22	Exit Short	Long	07/06/2022	29305	1	-1338.9	-4.79	-20620.5	-0.14	681.39	2.44	1693.4	6.06
22	Entry Short	Short	02/06/2022	27966	1	-1338.9	-4.79	-20620.5	-0.14	681.39	2.44	1693.4	6.06
21	Exit Long	Short	02/06/2022	27966	1	-1471.7	-5	-19281.64	-0.15	732.97	2.49	1926.37	6.54
21	Entry Long	Long	31/05/2022	29438	1	-1471.7	-5	-19281.64	-0.15	732.97	2.49	1926.37	6.54
20	Exit Short	Long	31/05/2022	29438	1	5230.41	15.09	-17809.94	0.54	10442.2	30.12	115.36	0.33
20	Entry Short	Short	06/05/2022	34669	1	5230.41	15.09	-17809.94	0.54	10442.2	30.12	115.36	0.33
19	Exit Long	Short	06/05/2022	34669	1	-2692.6	-7.21	-23040.35	-0.27	163.57	0.44	3544.65	9.49
19	Entry Long	Long	05/05/2022	37361	1	-2692.6	-7.21	-23040.35	-0.27	163.57	0.44	3544.65	9.49
18	Exit Short	Long	05/05/2022	37361	1	-740.51	-2.02	-20347.78	-0.08	1118.56	3.05	1111.74	3.04
18	Entry Short	Short	30/04/2022	36621	1	-740.51	-2.02	-20347.78	-0.08	1118.56	3.05	1111.74	3.04
17	Exit Long	Short	30/04/2022	36621	1	-1216.4	-3.21	-19607.27	-0.12	80.98	0.21	1646.92	4.35
17	Entry Long	Long	29/04/2022	37837	1	-1216.4	-3.21	-19607.27	-0.12	80.98	0.21	1646.92	4.35
16	Exit Short	Long	29/04/2022	37837	1	-2009.4	-5.61	-18390.91	-0.2	215.34	0.6	2635.61	7.36
16	Entry Short	Short	27/04/2022	35828	1	-2009.4	-5.61	-18390.91	-0.2	215.34	0.6	2635.61	7.36
15	Exit Long	Short	27/04/2022	35828	1	-1922.6	-5.09	-16381.53	-0.2	438.07	1.16	2312.12	6.12
15	Entry Long	Long	26/04/2022	37750	1	-1922.6	-5.09	-16381.53	-0.2	438.07	1.16	2312.12	6.12
14	Exit Short	Long	26/04/2022	37750	1	359.19	0.94	-14458.95	0.04	2543.31	6.67	1369.29	3.59
14	Entry Short	Short	21/04/2022	38109	1	359.19	0.94	-14458.95	0.04	2543.31	6.67	1369.29	3.59
13	Exit Long	Short	21/04/2022	38109	1	-330.51	-0.86	-14818.14	-0.03	538.62	1.4	730.94	1.9
13	Entry Long	Long	20/04/2022	38440	1	-330.51	-0.86	-14818.14	-0.03	538.62	1.4	730.94	1.9
12	Exit Short	Long	20/04/2022	38440	1	1173.21	2.96	-14487.63	0.12	3873.57	9.78	868.82	2.19
12	Entry Short	Short	07/04/2022	39613	1	1173.21	2.96	-14487.63	0.12	3873.57	9.78	868.82	2.19
11	Exit Long	Short	07/04/2022	39613	1	2306.83	6.18	-15660.84	0.23	6601.33	17.69	1026.17	2.75
11	Entry Long	Long	17/03/2022	37306	1	2306.83	6.18	-15660.84	0.23	6601.33	17.69	1026.17	2.75
10	Exit Short	Long	17/03/2022	37306	1	-1483.7	-4.14	-17967.67	-0.15	406.48	1.13	2177.5	6.08
10	Entry Short	Short	16/03/2022	35822	1	-1483.7	-4.14	-17967.67	-0.15	406.48	1.13	2177.5	6.08
9	Exit Long	Short	16/03/2022	35822	1	-444.6	-1.23	-16483.94	-0.05	164.76	0.45	1496.97	4.13
9	Entry Long	Long	15/03/2022	36267	1	-444.6	-1.23	-16483.94	-0.05	164.76	0.45	1496.97	4.13
8	Exit Short	Long	15/03/2022	36267	1	-452.06	-1.26	-16039.34	-0.05	1415.01	3.95	736.38	2.06
8	Entry Short	Short	11/03/2022	35815	1	-452.06	-1.26	-16039.34	-0.05	1415.01	3.95	736.38	2.06
7	Exit Long	Short	11/03/2022	35815	1	-2082.1	-5.49	-15587.28	-0.21	142.69	0.38	3017.06	7.96
7	Entry Long	Long	10/03/2022	37897	1	-2082.1	-5.49	-15587.28	-0.21	142.69	0.38	3017.06	7.96
6	Exit Short	Long	10/03/2022	37897	1	-2056.2	-5.74	-13505.23	-0.21	1576.13	4.4	2959.11	8.26
6	Entry Short	Short	05/03/2022	35841	1	-2056.2	-5.74	-13505.23	-0.21	1576.13	4.4	2959.11	8.26
5	Exit Long	Short	05/03/2022	35841	1	-2661.2	-6.91	-11449.06	-0.27	2395.02	6.22	3198.53	8.31
5	Entry Long	Long	01/03/2022	38502	1	-2661.2	-6.91	-11449.06	-0.27	2395.02	6.22	3198.53	8.31
4	Exit Short	Long	01/03/2022	38502	1	-2828.2	-7.93	-8787.83	-0.28	5129.74	14.38	3716.5	10.42
4	Entry Short	Short	18/02/2022	35674	1	-2828.2	-7.93	-8787.83	-0.28	5129.74	14.38	3716.5	10.42
3	Exit Long	Short	18/02/2022	35674	1	-647.88	-1.78	-5959.66	-0.07	3623.17	9.98	1071.83	2.95
3	Entry Long	Long	05/02/2022	36322	1	-647.88	-1.78	-5959.66	-0.07	3623.17	9.98	1071.83	2.95
2	Exit Short	Long	05/02/2022	36322	1	-3637.7	-11.1	-5311.78	-0.36	939.11	2.87	3777.09	11.56
2	Entry Short	Short	03/02/2022	32684	1	-3637.7	-11.1	-5311.78	-0.36	939.11	2.87	3777.09	11.56
1	Exit Long	Short	03/02/2022	32684	1	-1674.1	-4.87	-1674.07	-0.17	159.17	0.46	1958.19	5.7
1	Entry Long	Long	02/02/2022	34358	1	-1674.1	-4.87	-1674.07	-0.17	159.17	0.46	1958.19	5.7

Appendix 18. List of trades (Bollinger bands)

LIST OF TRADES (BOLLINGER)

Trade #	Type	Signal	Date/Time	Price EUR	Contracts	Profit EUR	Profit %	Cum. Profit EUR	Cum. Profit %	Run-up EUR	Run-up %	Drawdown EUR	Drawdown %
29	Exit Short	Close entry(s) order Short	19/12/2023	39058.5	1	1064.72	2.65	23625.71	0.1	2981.95	7.43	1459.45	3.64
29	Entry Short	Short	08/12/2023	40123.2	1	1064.72	2.65	23625.71	0.1	2981.95	7.43	1459.45	3.64
28	Exit Short	Close entry(s) order Short	16/11/2023	34956.8	1	-8090.1	-30.1	22560.99	-0.78	154.79	0.58	8607.58	32.04
28	Entry Short	Short	18/10/2023	26866.7	1	-8090.1	-30.1	22560.99	-0.78	154.79	0.58	8607.58	32.04
27	Exit Short	Close entry(s) order Short	17/10/2023	27016.4	1	-1556.8	-6.11	30651.1	-0.15	1044.57	4.1	3100.42	12.18
27	Entry Short	Short	21/09/2023	25459.6	1	-1556.8	-6.11	30651.1	-0.15	1044.57	4.1	3100.42	12.18
26	Exit Long	Close entry(s) order Long	31/08/2023	24982.1	1	1004.9	4.19	32207.93	0.1	1977.96	8.25	657.19	2.74
26	Entry Long	Long	22/08/2023	23977.2	1	1004.9	4.19	32207.93	0.1	1977.96	8.25	657.19	2.74
25	Exit Short	Long	22/08/2023	23977.2	1	2948.01	10.95	31203.03	0.29	3812.2	14.16	187.92	0.7
25	Entry Short	Short	10/08/2023	26925.2	1	2948.01	10.95	31203.03	0.29	3812.2	14.16	187.92	0.7
24	Exit Long	Close entry(s) order Long	03/08/2023	26651.2	1	-384.23	-1.42	28255.02	-0.04	258.99	0.96	977.37	3.62
24	Entry Long	Long	16/07/2023	27035.4	1	-384.23	-1.42	28255.02	-0.04	258.99	0.96	977.37	3.62
23	Exit Short	Close entry(s) order Short	14/07/2023	28026.7	1	-93.17	-0.33	28639.25	-0.01	929.79	3.33	1064.71	3.81
23	Entry Short	Short	26/06/2023	27933.5	1	-93.17	-0.33	28639.25	-0.01	929.79	3.33	1064.71	3.81
22	Exit Long	Short	26/06/2023	27933.5	1	4570.59	19.56	28732.42	0.45	5499.96	23.54	356.65	1.53
22	Entry Long	Long	16/06/2023	23362.9	1	4570.59	19.56	28732.42	0.45	5499.96	23.54	356.65	1.53
21	Exit Long	Close entry(s) order Long	08/06/2023	24591.3	1	-862.36	-3.39	24161.83	-0.08	135.54	0.53	1033.61	4.06
21	Entry Long	Long	07/06/2023	25453.6	1	-862.36	-3.39	24161.83	-0.08	135.54	0.53	1033.61	4.06
20	Exit Short	Close entry(s) order Short	03/06/2023	25452	1	358.68	1.39	25024.19	0.04	1162.76	4.5	135.09	0.52
20	Entry Short	Short	31/05/2023	25810.7	1	358.68	1.39	25024.19	0.04	1162.76	4.5	135.09	0.52
19	Exit Short	Close entry(s) order Short	28/04/2023	26737.4	1	471.4	1.73	24665.51	0.05	2776.77	10.21	841.29	3.09
19	Entry Short	Short	13/04/2023	27208.8	1	471.4	1.73	24665.51	0.05	2776.77	10.21	841.29	3.09
18	Exit Short	Close entry(s) order Short	05/04/2023	25719.4	1	220.99	0.85	24194.11	0.02	1390.35	5.36	981.24	3.78
18	Entry Short	Short	21/03/2023	25940.4	1	220.99	0.85	24194.11	0.02	1390.35	5.36	981.24	3.78
17	Exit Long	Short	21/03/2023	25940.4	1	6693.55	34.78	23973.12	0.66	7547.2	39.21	138.53	0.72
17	Entry Long	Long	12/03/2023	19246.8	1	6693.55	34.78	23973.12	0.66	7547.2	39.21	138.53	0.72
16	Exit Short	Close entry(s) order Short	02/03/2023	22152.1	1	-106.41	-0.48	17279.57	-0.01	445.72	2.02	1613.58	7.32
16	Entry Short	Short	17/02/2023	22045.7	1	-106.41	-0.48	17279.57	-0.01	445.72	2.02	1613.58	7.32
15	Exit Long	Short	17/02/2023	22045.7	1	1548.11	7.55	17385.98	0.15	3152.39	15.38	550.14	2.68
15	Entry Long	Long	12/02/2023	20497.6	1	1548.11	7.55	17385.98	0.15	3152.39	15.38	550.14	2.68
14	Exit Short	Long	12/02/2023	20497.6	1	-903.5	-4.61	15837.87	-0.09	814.11	4.15	2517.89	12.85
14	Entry Short	Short	18/01/2023	19594.1	1	-903.5	-4.61	15837.87	-0.09	814.11	4.15	2517.89	12.85
13	Exit Long	Short	18/01/2023	19594.1	1	3692	23.22	16741.37	0.36	3989.84	25.09	598.85	3.77
13	Entry Long	Long	21/12/2022	15902.1	1	3692	23.22	16741.37	0.36	3989.84	25.09	598.85	3.77
12	Exit Short	Long	21/12/2022	15902.1	1	767.99	4.61	13049.37	0.08	1335.65	8.01	48.74	0.29
12	Entry Short	Short	15/12/2022	16670.1	1	767.99	4.61	13049.37	0.08	1335.65	8.01	48.74	0.29
11	Exit Long	Short	15/12/2022	16670.1	1	483.83	2.99	12281.38	0.05	1013.73	6.26	1076.27	6.65
11	Entry Long	Long	13/11/2022	16186.3	1	483.83	2.99	12281.38	0.05	1013.73	6.26	1076.27	6.65
10	Exit Short	Long	13/11/2022	16186.3	1	4168.7	20.48	11797.55	0.41	4803.96	23.6	1207.38	5.93
10	Entry Short	Short	28/10/2022	20355	1	4168.7	20.48	11797.55	0.41	4803.96	23.6	1207.38	5.93
9	Exit Long	Close entry(s) order Long	14/09/2022	20199.8	1	-1227.6	-5.73	7628.85	-0.12	1024.42	4.78	2672.35	12.47
9	Entry Long	Long	22/08/2022	21427.4	1	-1227.6	-5.73	7628.85	-0.12	1024.42	4.78	2672.35	12.47
8	Exit Short	Close entry(s) order Short	28/07/2022	22506.8	1	295.51	1.3	8856.45	0.03	2421.21	10.62	471.66	2.07
8	Entry Short	Short	21/07/2022	22802.3	1	295.51	1.3	8856.45	0.03	2421.21	10.62	471.66	2.07
7	Exit Short	Close entry(s) order Short	14/07/2022	20154.6	1	1043.93	4.92	8560.94	0.1	2320.57	10.95	12.09	0.06
7	Entry Short	Short	10/07/2022	21198.6	1	1043.93	4.92	8560.94	0.1	2320.57	10.95	12.09	0.06
6	Exit Long	Short	10/07/2022	21198.6	1	1721.7	8.84	7517.01	0.17	2601.81	13.36	2700.57	13.87
6	Entry Long	Long	18/06/2022	19476.9	1	1721.7	8.84	7517.01	0.17	2601.81	13.36	2700.57	13.87
5	Exit Long	Close entry(s) order Long	02/06/2022	27966.4	1	132.17	0.47	5795.31	0.01	2363.54	8.49	1668.21	5.99
5	Entry Long	Long	13/05/2022	27834.2	1	132.17	0.47	5795.31	0.01	2363.54	8.49	1668.21	5.99
4	Exit Long	Close entry(s) order Long	30/04/2022	36620.6	1	-408.3	-1.1	5663.14	-0.04	2449.74	6.62	1590.86	4.3
4	Entry Long	Long	13/04/2022	37028.9	1	-408.3	-1.1	5663.14	-0.04	2449.74	6.62	1590.86	4.3
3	Exit Short	Long	13/04/2022	37028.9	1	3525.46	8.69	6071.44	0.35	4554.32	11.23	3353.21	8.27
3	Entry Short	Short	27/03/2022	40554.3	1	3525.46	8.69	6071.44	0.35	4554.32	11.23	3353.21	8.27
2	Exit Short	Close entry(s) order Short	01/03/2022	38502.1	1	393.22	1.01	2545.98	0.04	8351.13	21.47	1049.66	2.7
2	Entry Short	Short	10/02/2022	38895.3	1	393.22	1.01	2545.98	0.04	8351.13	21.47	1049.66	2.7
1	Exit Long	Short	10/02/2022	38895.3	1	2152.76	5.86	2152.76	0.22	3149.49	8.57	7582.67	20.64
1	Entry Long	Long	09/01/2022	36742.6	1	2152.76	5.86	2152.76	0.22	3149.49	8.57	7582.67	20.64

Appendix 19. List of trades (Stochastic Oscillator)

LIST OF TRADES (STOCHASTIC OSC.)

Trade #	Type	Signal	Date/Time	Price EUF	Contracts	Profit EUR	Profit %	Cum. Profit EUF	Cum. Profit %	Run-up EUF	Run-up %	Drawdown EUR	Drawdown %
40	Exit Short	Exit Short	07/12/2023	40756.4	1	-90.91	-0.22	842.86	-0.01	0	0	90.91	0.22
40	Entry Short	Short	07/12/2023	40665.5	1	-90.91	-0.22	842.86	-0.01	0	0	90.91	0.22
39	Exit Short	Exit Short	03/11/2023	32732.6	1	205.97	0.63	933.77	0.02	205.97	0.63	0	0
39	Entry Short	Short	03/11/2023	32938.6	1	205.97	0.63	933.77	0.02	205.97	0.63	0	0
38	Exit Short	Exit Short	29/10/2023	32380.1	1	-111.84	-0.35	727.8	-0.01	123.55	0.38	111.84	0.35
38	Entry Short	Short	29/10/2023	32268.2	1	-111.84	-0.35	727.8	-0.01	123.55	0.38	111.84	0.35
37	Exit Short	Exit Short	25/10/2023	31817	1	190.46	0.6	839.64	0.02	190.46	0.6	0	0
37	Entry Short	Short	25/10/2023	32007.5	1	190.46	0.6	839.64	0.02	190.46	0.6	0	0
36	Exit Short	Exit Short	03/10/2023	26344.3	1	-105.81	-0.4	649.18	-0.01	0	0	105.81	0.4
36	Entry Short	Short	03/10/2023	26238.5	1	-105.81	-0.4	649.18	-0.01	0	0	105.81	0.4
35	Exit Short	Exit Short	22/09/2023	25014.6	1	-96.62	-0.39	754.99	-0.01	73.82	0.3	96.62	0.39
35	Entry Short	Short	22/09/2023	24918	1	-96.62	-0.39	754.99	-0.01	73.82	0.3	96.62	0.39
34	Exit Long	Exit Long	21/07/2023	26964.3	1	187.27	0.7	851.61	0.02	187.27	0.7	68.68	0.26
34	Entry Long	Long	21/07/2023	26777	1	187.27	0.7	851.61	0.02	187.27	0.7	68.68	0.26
33	Exit Short	Exit Short	03/07/2023	28146.8	1	-96.4	-0.34	664.34	-0.01	19.69	0.07	96.4	0.34
33	Entry Short	Short	03/07/2023	28050.4	1	-96.4	-0.34	664.34	-0.01	19.69	0.07	96.4	0.34
32	Exit Short	Exit Short	01/07/2023	28009.6	1	-98.44	-0.35	760.74	-0.01	134.81	0.48	98.44	0.35
32	Entry Short	Short	01/07/2023	27911.2	1	-98.44	-0.35	760.74	-0.01	134.81	0.48	98.44	0.35
31	Exit Short	Exit Short	27/06/2023	27835.3	1	-102.47	-0.37	859.18	-0.01	79.04	0.29	102.47	0.37
31	Entry Short	Short	27/06/2023	27732.9	1	-102.47	-0.37	859.18	-0.01	79.04	0.29	102.47	0.37
30	Exit Short	Exit Short	25/06/2023	27814.6	1	207.61	0.74	961.65	0.02	207.61	0.74	0	0
30	Entry Short	Short	25/06/2023	28022.2	1	207.61	0.74	961.65	0.02	207.61	0.74	0	0
29	Exit Long	Exit Long	16/06/2023	23264.9	1	-98.06	-0.42	754.04	-0.01	0	0	98.06	0.42
29	Entry Long	Long	16/06/2023	23362.9	1	-98.06	-0.42	754.04	-0.01	0	0	98.06	0.42
28	Exit Long	Exit Long	25/04/2023	24789.3	1	-98.68	-0.4	852.1	-0.01	0	0	98.68	0.4
28	Entry Long	Long	25/04/2023	24888	1	-98.68	-0.4	852.1	-0.01	0	0	98.68	0.4
27	Exit Short	Exit Short	14/04/2023	27327.6	1	200.06	0.73	950.78	0.02	200.06	0.73	0	0
27	Entry Short	Short	14/04/2023	27527.6	1	200.06	0.73	950.78	0.02	200.06	0.73	0	0
26	Exit Short	Exit Short	23/03/2023	25243.7	1	-110.21	-0.44	750.72	-0.01	154.43	0.61	110.21	0.44
26	Entry Short	Short	23/03/2023	25133.5	1	-110.21	-0.44	750.72	-0.01	154.43	0.61	110.21	0.44
25	Exit Short	Exit Short	21/03/2023	25727.8	1	212.59	0.82	860.93	0.02	212.59	0.82	0	0
25	Entry Short	Short	21/03/2023	25940.4	1	212.59	0.82	860.93	0.02	212.59	0.82	0	0
24	Exit Long	Exit Long	12/03/2023	19151.9	1	-94.93	-0.49	648.34	-0.01	0	0	94.93	0.49
24	Entry Long	Long	12/03/2023	19246.8	1	-94.93	-0.49	648.34	-0.01	0	0	94.93	0.49
23	Exit Long	Exit Long	08/03/2023	20932.6	1	-101.19	-0.48	743.27	-0.01	85.2	0.41	101.19	0.48
23	Entry Long	Long	08/03/2023	21033.8	1	-101.19	-0.48	743.27	-0.01	85.2	0.41	101.19	0.48
22	Exit Long	Exit Long	14/02/2023	20208.5	1	-100.33	-0.49	844.46	-0.01	0	0	100.33	0.49
22	Entry Long	Long	14/02/2023	20308.9	1	-100.33	-0.49	844.46	-0.01	0	0	100.33	0.49
21	Exit Short	Exit Short	31/01/2023	21147.1	1	-103.77	-0.49	944.79	-0.01	103.64	0.49	103.77	0.49
21	Entry Short	Short	31/01/2023	21043.3	1	-103.77	-0.49	944.79	-0.01	103.64	0.49	103.77	0.49
20	Exit Short	Exit Short	24/01/2023	21188.3	1	-99.41	-0.47	1048.56	-0.01	0	0	99.41	0.47
20	Entry Short	Short	24/01/2023	21088.9	1	-99.41	-0.47	1048.56	-0.01	0	0	99.41	0.47
19	Exit Short	Exit Short	17/01/2023	19368.4	1	200.66	1.03	1147.97	0.02	200.66	1.03	0	0
19	Entry Short	Short	17/01/2023	19569.1	1	200.66	1.03	1147.97	0.02	200.66	1.03	0	0
18	Exit Short	Exit Short	15/01/2023	19127.4	1	205.8	1.06	947.31	0.02	205.8	1.06	66.76	0.35
18	Entry Short	Short	15/01/2023	19333.2	1	205.8	1.06	947.31	0.02	205.8	1.06	66.76	0.35
17	Exit Short	Exit Short	10/01/2023	16104.8	1	-93.62	-0.58	741.51	-0.01	24.75	0.15	93.62	0.58
17	Entry Short	Short	10/01/2023	16011.2	1	-93.62	-0.58	741.51	-0.01	24.75	0.15	93.62	0.58
16	Exit Long	Exit Long	21/12/2022	15809.7	1	-92.45	-0.58	835.13	-0.01	39.34	0.25	92.45	0.58
16	Entry Long	Long	21/12/2022	15902.1	1	-92.45	-0.58	835.13	-0.01	39.34	0.25	92.45	0.58
15	Exit Long	Exit Long	24/11/2022	15828.8	1	-88.11	-0.55	927.58	-0.01	0	0	88.11	0.55
15	Entry Long	Long	24/11/2022	15916.9	1	-88.11	-0.55	927.58	-0.01	0	0	88.11	0.55
14	Exit Long	Exit Long	20/11/2022	16054.6	1	-100.13	-0.62	1015.69	-0.01	50.61	0.31	100.13	0.62
14	Entry Long	Long	20/11/2022	16154.7	1	-100.13	-0.62	1015.69	-0.01	50.61	0.31	100.13	0.62
13	Exit Long	Exit Long	16/11/2022	16202.1	1	-103.52	-0.63	1115.82	-0.01	81.6	0.5	103.52	0.63

13	Entry Long	Long	16/11/2022	16305.6	1	-103.52	-0.63	1115.82	-0.01	81.6	0.5	103.52	0.63
12	Exit Long	Exit Long	31/08/2022	19975.5	1	197.73	1	1219.34	0.02	197.73	1	11.34	0.06
12	Entry Long	Long	31/08/2022	19777.8	1	197.73	1	1219.34	0.02	197.73	1	11.34	0.06
11	Exit Long	Exit Long	23/08/2022	21433.2	1	-102.36	-0.48	1021.61	-0.01	183.06	0.85	102.36	0.48
11	Entry Long	Long	23/08/2022	21535.5	1	-102.36	-0.48	1021.61	-0.01	183.06	0.85	102.36	0.48
10	Exit Short	Exit Short	22/07/2022	22446.1	1	207.26	0.91	1123.97	0.02	207.26	0.91	0	0
10	Entry Short	Short	22/07/2022	22653.4	1	207.26	0.91	1123.97	0.02	207.26	0.91	0	0
9	Exit Long	Exit Long	20/06/2022	19791.5	1	210.05	1.07	916.71	0.02	210.05	1.07	0	0
9	Entry Long	Long	20/06/2022	19581.4	1	210.05	1.07	916.71	0.02	210.05	1.07	0	0
8	Exit Long	Exit Long	16/06/2022	21782.9	1	211	0.98	706.66	0.02	211	0.98	0	0
8	Entry Long	Long	16/06/2022	21571.9	1	211	0.98	706.66	0.02	211	0.98	0	0
7	Exit Long	Exit Long	12/05/2022	27786.5	1	205.15	0.74	495.66	0.02	205.15	0.74	0	0
7	Entry Long	Long	12/05/2022	27581.4	1	205.15	0.74	495.66	0.02	205.15	0.74	0	0
6	Exit Long	Exit Long	19/04/2022	37759.2	1	-105.33	-0.28	290.51	-0.01	0	0	105.33	0.28
6	Entry Long	Long	19/04/2022	37864.5	1	-105.33	-0.28	290.51	-0.01	0	0	105.33	0.28
5	Exit Long	Exit Long	13/04/2022	36930.4	1	-98.45	-0.27	395.84	-0.01	0	0	98.45	0.27
5	Entry Long	Long	13/04/2022	37028.9	1	-98.45	-0.27	395.84	-0.01	0	0	98.45	0.27
4	Exit Short	Exit Short	29/03/2022	42679	1	190.43	0.44	494.29	0.02	190.43	0.44	0	0
4	Entry Short	Short	29/03/2022	42869.5	1	190.43	0.44	494.29	0.02	190.43	0.44	0	0
3	Exit Long	Exit Long	23/02/2022	33973.7	1	191.41	0.57	303.86	0.02	191.41	0.57	0	0
3	Entry Long	Long	23/02/2022	33782.2	1	191.41	0.57	303.86	0.02	191.41	0.57	0	0
2	Exit Short	Exit Short	09/02/2022	38723.3	1	-97.56	-0.25	112.45	-0.01	0	0	97.56	0.25
2	Entry Short	Short	09/02/2022	38625.7	1	-97.56	-0.25	112.45	-0.01	0	0	97.56	0.25
1	Exit Long	Exit Long	10/01/2022	37102.7	1	210.01	0.57	210.01	0.02	210.01	0.57	0	0
1	Entry Long	Long	10/01/2022	36892.7	1	210.01	0.57	210.01	0.02	210.01	0.57	0	0

Appendix 20. List of trades (Relative Strength Index)

LIST OF TRADES (RSI)

Trade #	Type	Signal	Date/Time	Price EUF	Contracts	Profit EUR	Profit %	Cum. Profit EUF	Cum. Profit %	Run-up EUF	Run-up %	Drawdown EUR	Drawdown %
4	Exit Long	Close entry(s) order Long	04/11/2023	32403.9	1	8063.28	33.13	11352.88	0.8	9622.51	39.53	1165.57	4.79
4	Entry Long	Long	24/08/2023	24340.6	1	8063.28	33.13	11352.88	0.8	9622.51	39.53	1165.57	4.79
3	Exit Long	Close entry(s) order Long	23/03/2023	25133.5	1	5886.73	30.59	3289.6	0.59	7576.03	39.36	138.53	0.72
3	Entry Long	Long	12/03/2023	19246.8	1	5886.73	30.59	3289.6	0.59	7576.03	39.36	138.53	0.72
2	Exit Long	Close entry(s) order Long	31/01/2023	21043.3	1	-7833.1	-27.1	-2597.13	-0.78	1321.34	4.58	13766.42	47.67
2	Entry Long	Long	15/05/2022	28876.4	1	-7833.1	-27.1	-2597.13	-0.78	1321.34	4.58	13766.42	47.67
1	Exit Long	Close entry(s) order Long	31/03/2022	42128.7	1	5235.97	14.19	5235.97	0.52	7014.81	19.01	7732.81	20.96
1	Entry Long	Long	10/01/2022	36892.7	1	5235.97	14.19	5235.97	0.52	7014.81	19.01	7732.81	20.96